When Explanations Outlive Their Data: Faithfulness Decoupling in Graph-Attention MARL Fleet Dispatch under Telemetry Degradation

A Master's Dissertation

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By

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DECLARATION

I declare that this dissertation is my own original work carried out under the supervision of Farhan S. Ujager and that all sources and references have been acknowledged appropriately. This dissertation has not been submitted previously for academic credit at De Montfort University, Dubai, or any other institution.

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Date: ______________________________

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LIST OF ABBREVIATIONS

Abbreviation	Full form
AoI	Age of Information
CI	Confidence Interval
CTDE	Centralized Training with Decentralized Execution
DEF	Decision-level Explanation Faithfulness
GAT	Graph Attention Network
GNN	Graph Neural Network
GxI	Gradient x Input
LOO	Leave-One-Out
MAPPO	Multi-Agent Proximal Policy Optimization
MARL	Multi-Agent Reinforcement Learning
MLP	Multilayer Perceptron
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
SUMO	Simulation of Urban Mobility
WAMSN	Weighted Attention Mass on Stale Nodes
XAI	Explainable Artificial Intelligence
XRL	Explainable Reinforcement Learning

ABSTRACT

Graph-attention multi-agent reinforcement learning (GAT-MARL) can expose attention weights for use as candidate explanations of fleet-dispatch decisions. An attention map can remain visible when vehicle telemetry becomes stale, but it does not show whether its evidence is current or decision-relevant. This dissertation asks: **can raw graph-attention weights be trusted as explanations when fleet telemetry is degraded?**

The experiment simulates 20 taxis in SUMO. Tunnel entry triggers a simulated outage, and the observation layer freezes the affected vehicle's last-known position and speed for 10, 20, 30, or 60 seconds. SUMO retains the true state, allowing paired clean and degraded observations. GAT policies trained on clean observations and on 30-second outage observations are evaluated across three independent training seeds. Decision-level explanation faithfulness (DEF) compares highly attended node subsets with type-matched random subsets, while WAMSN measures stale-data exposure. A construct-validity audit controls for request nodes that also define available actions.

Under stochastic action sampling, all selected policies exceed legal-random and naive greedy lower bounds on held-out demand. A deterministic diagnostic, however, produces no pickups for any GAT checkpoint. The reported capability therefore belongs to the sampled policy distribution, not to a reliable argmax dispatcher.

The results do not validate raw attention as a dependable explanation. Of the 56,550 decisions with at least one available request, 93.0% select no-op. Action-stratified analysis shows that the main duration and exposure associations are not reproduced for dispatch actions. In four of six checkpoints, the paired DEF direction also differs between the all-decision and dispatch strata. Under clean telemetry, DEF varies in sign across checkpoints and is lower than a leave-one-out perturbation control in all six selected checkpoints. Longer outages increase stale-data exposure, but the paired change in attention toward stale nodes is positive for two training seeds and negative for one under both training regimes. Within-policy tests show an association between greater exposure and lower DEF, but the direct paired DEF change is negative for four checkpoints and positive for two. The result also depends on the selected checkpoint and the rule used to aggregate attention into a single explanation. A random-trigger control retains the attention-shift direction in five of six checkpoints and the DEF-shift direction in four of six, so the variation cannot be attributed only to the fixed tunnel location.

The conclusion is not that stale telemetry always reduces faithfulness. It is that raw attention provides no stable guarantee of either data freshness or decision relevance across independently trained policies or action types. An attention map can therefore outlive the data supporting it while still appearing complete. In response, the dissertation proposes a freshness-aware explanation assurance approach that separates freshness monitoring from attention, applies action-aware faithfulness tests, checks aggregation sensitivity, and repeats the audit for each released checkpoint. Attention weights should remain model internals unless these checks give consistent results across independent training runs.

Keywords: explainable reinforcement learning; graph attention; multi-agent reinforcement learning; fleet dispatch; telemetry degradation; Age of Information; faithfulness.

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Chapter 1: Introduction

1.1 Context

Fleet dispatch is relational. A taxi's useful action depends on nearby vehicles, open passenger requests, road conditions, and competition for the same demand. Graph neural networks represent these relationships directly, and graph attention networks (GATs) add learned weights over neighboring nodes (Scarselli et al., 2009; Velickovic et al., 2018). Because the weights are easy to display, they are often read as an explanation of which vehicles or requests influenced a decision.

That reading is convenient, but it is not automatically valid. Prior work has shown that an attention weight can be weakly related to the evidence that actually changes a model's output (Jain and Wallace, 2019; Serrano and Smith, 2019; Wiegreffe and Pinter, 2019; Liu et al., 2022). The issue becomes more serious in an operational system because the input itself may no longer describe the current world.

1.2 Problem statement

Vehicle telemetry can be interrupted by tunnels and other signal-loss areas. During an interruption, a dispatch platform may keep the last received position and speed. The reading remains available, but its Age of Information (AoI) grows (Kaul, Yates and Gruteser, 2012). An attention map can therefore look complete even though some nodes represent old information. The interface reveals where the model placed attention, but it does not reveal whether that information is fresh or whether removing it would change the decision.

This creates an assurance problem. An operator may read a complete attention map as a trustworthy reason even when its evidence is stale, and stable dispatch performance cannot show whether the explanation is valid. The dissertation is therefore not trying to improve the number of pickups. It is trying to determine whether raw graph-attention weights provide dependable evidence about a decision under clean and degraded telemetry.

This dissertation uses **faithfulness decoupling** to describe the risk that an explanation remains available and visually plausible after its supporting data have become stale, without a dependable link between the displayed weights and the evidence that changes the decision. One possible pattern is declining faithfulness while dispatch performance remains stable. The broader problem, however, is whether attention provides reliable assurance under the tested conditions.

1.3 Research questions

The central research question is:

Can raw graph-attention weights be trusted as explanations of fleet-dispatch decisions when vehicle telemetry becomes stale?

Four operational questions provide the evidence needed to answer it:

1. **RQ1:** Is graph attention a faithful explanation under clean telemetry?
2. **RQ2:** When tunnel-triggered outages occur, does attention move toward stale vehicle nodes?

3. **RQ3:** As outage duration increases, does the relationship between attention, stale-data exposure, and decision relevance remain dependable?
4. **RQ4:** Does training the policy under telemetry degradation make the explanation response more reliable?

The construct-validity audit supports RQ1-RQ3. It is not a separate primary problem. Its purpose is to check that the faithfulness metric measures hidden information rather than the accidental deletion of an available action.

1.4 Objectives

The measurable objectives are to:

- build a reproducible SUMO fleet-dispatch benchmark with training, validation, and held-out test demand;
- train clean and degradation-aware policies across three independent seeds;
- trigger signal loss from tunnel entry while applying the resulting freeze at the observation boundary;
- compare clean telemetry with fixed observation-layer outages of 10, 20, 30, and 60 seconds;
- measure explanation faithfulness with action-aware, type-matched counterfactual controls;
- check whether the faithfulness evaluator is sensitive to an LOO perturbation ranking and quantify the resolution lost through top-k overlap;
- measure how much attention is assigned to stale vehicle information;
- test whether conclusions change across attention layers, heads, rollout, and the query row used as the explanation;
- treat the trained policy, rather than each individual decision, as the unit for judging whether a finding is consistent.

1.5 Main findings

The experiment gives one consistent overall answer: **raw attention weights are not validated as dependable explanations by this study**. This is a negative assurance result, supported by three connected findings.

First, clean type-matched DEF varies from -0.0009 to +0.0031 across the six selected checkpoints. An LOO perturbation control is larger and positive in every checkpoint. Taxi-only rank agreement with LOO is positive but ranges from 0.088 to 0.699. Attention therefore shows some agreement with a decision-relevant perturbation ranking, but the strength of that agreement is not reproducible across checkpoints. The supporting audit also shows why a uniform random baseline is misleading when request nodes are actions.

Second, longer outages consistently increase WAMSN. This result appears in all three GAT checkpoints and all three GAT-Outage checkpoints. It shows that more of the displayed attention is attached to stale information when stale exposure lasts longer. It does **not** show that AoI causes lower faithfulness.

Third, the paired change in stale-node attention is not consistent across training seeds. It is positive for checkpoints trained with seeds 42 and 43 but negative for checkpoints trained with seed 44 in both GAT and GAT-Outage. H1 is supported in five of six selected checkpoints, and the within-episode WAMSN-DEF association is supported in all six. However, the direct clean-twin DEF shift is negative for four checkpoints and positive for two. This distinction matters: a within-policy association can be reproducible without giving every trained policy the same response to degradation.

A 30-second random-trigger sensitivity check retains the attention-shift direction in five of six checkpoints and the probability-DEF direction in four of six. This reduces concern that the checkpoint pattern is only an artifact of the fixed tunnel location, while preserving the need for a cautious conclusion.

The action audit narrows this result further. After decisions with no available request are removed, 93.0% of records still select no-op and only 7.0% dispatch a request. The H1 and H4 patterns found in the combined data are not reproduced within the smaller dispatch stratum. In four of six checkpoints, the sign of the paired DEF change for dispatch decisions also differs from the sign in the combined analysis. The combined result therefore describes the sampled policy distribution; it cannot be presented as a dispatch-action-specific effect.

The result is also sensitive to how attention is turned into one explanation. Individual heads can reverse the stale-attention direction within a checkpoint, and changing from the self row to the selected request row mainly improves the checkpoints trained with seed 43. These checks define more precisely why the study does not establish a reliable default explanation.

Finally, degradation-aware training does not remove training-seed variation. All selected policies pass the declared stochastic capability and training stability gates. However, a deterministic diagnostic gives zero pickups for all six GAT checkpoints over three held-out episodes each. The learned action distributions can produce useful sampled behavior, but their argmax actions are collapsed to no-op. This limitation narrows the setting in which the explanation response can be interpreted. Taken together, the results do not prove that attention is always unfaithful. They show that attention cannot be assumed trustworthy by default; freshness and faithfulness must be checked separately for every released policy.

1.6 Contributions

The dissertation contributes:

- a paired clean/degraded benchmark in which SUMO ground truth is separated from the observation received by the policy;
- an explicit distinction between a tunnel trigger and an observation-layer outage;
- an action-aware faithfulness protocol for graphs where request nodes also define available actions, supported by LOO and overlap diagnostics;
- a sensitivity audit showing how layer, head, rollout, and query-row choices affect the reported attention explanation;
- evidence across independently trained policies that stale-data exposure is consistent, while attention reallocation and faithfulness effects are not;
- an action-stratified diagnostic showing that combined decision statistics do not reliably represent the smaller set of dispatch actions;

- a freshness-aware explanation assurance approach that turns the findings into release checks for telemetry freshness, action-aware faithfulness, attention aggregation, and independent training runs;
- a reproducible experiment runner, validation-selected checkpoints, preflight checks, machine-readable results, and training-seed synthesis.

1.7 Dissertation structure

Chapter 2 reviews MARL dispatch, graph attention, explanation faithfulness, and AoI. Chapter 3 describes the simulator, models, degradation layer, metrics, and statistical design. Chapter 4 reports the results. Chapter 5 discusses what can and cannot be concluded. Chapter 6 closes with the practical implications for trustworthy fleet-dispatch explanations.

Chapter 2: Literature Review

This chapter develops the research gap in a sequence of connected stages. It first explains why fleet dispatch is a relational reinforcement-learning problem. It then asks what an explanation should provide in sequential decision making, why raw attention is disputed as an explanation, how graph predictions are commonly explained, and why faithfulness tests can themselves be misleading. The final sections connect these issues to stale telemetry and position the experiment.

2.1 Reinforcement learning for fleet dispatch

Fleet dispatch links vehicles, passenger requests, and a road network over time. A decision for one taxi changes the demand available to others, so a collection of independent single-agent policies can create competition or duplicated assignments. Lin et al. (2018) model city-scale fleet management as a cooperative multi-agent reinforcement-learning (MARL) problem. Qin, Zhu and Ye (2022) show that reinforcement learning is used for matching, repositioning, pricing, and route decisions, often under changing supply and demand.

MARL offers several ways to represent coordination. MADDPG learns decentralized actors with centralized critics (Lowe et al., 2017). QMIX factorizes a team value into agent values while preserving a monotonic relation to the joint action (Rashid et al., 2020). MAPPO uses the more familiar PPO objective with centralized training and decentralized execution; its empirical stability makes it a practical baseline for cooperative tasks (Yu et al., 2022). This dissertation adopts MAPPO because the research focus is the explanation attached to each taxi's local decision, not a new MARL algorithm.

Graphs provide a natural representation of the local dispatch problem. Nodes can represent the acting taxi, nearby taxis, and open requests; edges permit their information to be combined (Scarselli et al., 2009). A graph attention network (GAT) assigns learned weights while aggregating neighboring nodes (Velickovic et al., 2018). Actor-Attention-Critic similarly uses attention for communication in MARL (Iqbal and Sha, 2019). This makes GAT-MARL a suitable test case, not because it is assumed to be the best dispatch model, but because it matches the relational, multi-agent structure of fleet dispatch while exposing an attention channel that can be audited. These weights are easy to visualize, which makes them tempting to present as reasons for an action. Ease of display, however, is not evidence that the weights identify the information that caused the decision.

Recent dispatch methods continue to use relational structure. BMG-Q represents ride-pooling dispatch through a local bipartite matching graph (Hu, Feng and Li, 2025); CoopRide studies cooperation across city grids (Wang et al., 2025); and DualG-MARL combines vehicle-state and task graphs (Sha et al., 2026). Their primary outcomes are dispatch or scheduling quality. They do not test whether graph weights remain valid explanations when vehicle telemetry becomes stale. These studies are therefore useful task benchmarks, but not direct faithfulness benchmarks.

2.2 Explainability in reinforcement learning

Explainable reinforcement learning (XRL) covers more than a visual saliency map. An explanation may describe which state features mattered, summarize a policy, contrast the selected action with an alternative, or show the sequence of events that led to an outcome. Reviews by Heuillet, Couthouis and Diaz-Rodriguez (2021), Puiutta and Veith (2020), and Milani et al. (2024) all stress that the method must match the user's question. A model developer debugging a policy and an operator checking one live action do not need the same output. The systematic taxonomy by Bekkemoen (2024) reaches the same conclusion across a larger body of recent XRL studies.

This dissertation concerns a local, post-decision operator question: *which visible vehicles or requests supported this dispatch action?* Earlier XRL work demonstrates several possible answers. Greydanus et al. (2018) perturb image regions to visualize what changes an Atari policy. Madumal et al. (2020) use a causal model to generate contrastive explanations. Mott et al. (2019) introduce an attention bottleneck intended to expose what an RL agent uses. These approaches differ in mechanism, but they share an important lesson: a useful picture and a faithful account of model dependence are separate properties.

That distinction is especially important in an operational dashboard. A plausible explanation may be easy for a person to read but weakly connected to the policy computation. A faithful explanation should change when information that matters to the action is removed or retained. The experiment therefore does not ask whether users like the attention map. It asks whether its ranking passes a controlled decision-level faithfulness test.

2.3 Is attention an explanation?

The modern debate began with Jain and Wallace (2019), who found that attention weights could be weakly correlated with other importance measures and that very different attention distributions could produce similar predictions. Wiegreffe and Pinter (2019) argued that this does not justify rejecting all attention explanations and proposed more careful tests. Serrano and Smith (2019) likewise showed that removing high-attention items can affect outputs, but that the relationship varies by model and task. The useful conclusion is not a universal yes or no. It is that attention needs an explicit claim and an appropriate test.

Attention became widely visible through Transformer models (Vaswani et al., 2017), but visibility also encouraged a broader interpretability claim than the mechanism alone can support. Lipton (2018) warns that the word *interpretability* often combines several different goals that should be evaluated separately.

Jacovi and Goldberg (2020) separate *plausibility*, which concerns whether an explanation looks reasonable to a person, from *faithfulness*, which concerns whether it reflects the model's actual reasoning process. This dissertation follows that distinction. The attention weights are not rejected because they are internal values, and they are not accepted because they are coupled to the actor. They are treated as a candidate explanation whose decision relevance must be measured.

Layer and head aggregation add another difficulty. A two-layer, multi-head network does not produce one unique attention map. Averaging heads, selecting a layer, taking a maximum, or composing attention across layers can produce different rankings. Abnar and Zuidema (2020) propose attention rollout and flow to account for information mixing across layers, and report stronger correlations with ablation and gradient importance than raw attention in Transformers. Although their model is not a GAT-MARL dispatcher, the methodological point transfers: an attention result is incomplete unless the aggregation rule is defined and its sensitivity is checked.

2.4 Explaining graph neural networks

GNN explanations may identify important features, nodes, edges, or subgraphs. GNNExplainer learns a compact subgraph and feature mask that preserves a prediction (Ying et al., 2019). PGExplainer parameterizes explanation generation so that it can be shared across examples (Luo et al., 2020). SubgraphX searches for important subgraphs using Shapley-value approximations (Yuan et al., 2021). These methods show that graph explanation is a distinct problem: removing one node can alter both its own features and the messages available to other nodes.

The survey by Yuan et al. (2023) organizes GNN explainers by target, mechanism, and scope. GraphFramEx goes further by comparing explanation methods under different user needs and combining necessity and sufficiency views (Amara et al., 2022). Its results show that no single method dominates every evaluation dimension. This supports the use of more than one diagnostic in the present study: DEF tests counterfactual relevance, taxi-only rank correlation compares attention with leave-one-out effects, and stale-attention measures describe freshness exposure.

Most GNN explainability benchmarks study node or graph classification. Fleet dispatch differs because some graph nodes also define actions. A passenger request is both information and a selectable action, while a nearby taxi is context only. Deleting these node types is therefore not the same intervention. This action-linked graph structure motivates the construct-validity audit in Section 3.7.

2.5 Faithfulness evaluation and its pitfalls

Comprehensiveness and sufficiency are common perturbation measures. In the ERASER benchmark, comprehensiveness asks how much a prediction weakens when the explanation is removed, while sufficiency asks how well the selected evidence alone preserves it (DeYoung et al., 2020). Liu et al. (2022) similarly test attention by measuring faithfulness violations. These measures are useful because they connect an explanation to model behavior instead of visual appeal.

Model-agnostic methods such as LIME also use local perturbations to estimate which inputs support a prediction (Ribeiro, Singh and Guestrin, 2016). Their usefulness depends on whether the perturbed samples remain meaningful for the model and task.

Perturbation is not automatically valid. Removing input features may create samples unlike the data seen during training. ROAR addresses this issue by removing features and retraining the model, showing why a simple deletion test can confound attribution quality with distribution shift (Hooker et al., 2019). Adebayo et al. (2018) provide a different sanity check: an explanation should respond when model parameters or training labels are randomized. Alvarez-Melis and Jaakkola (2018) show that explanation methods can also be locally unstable. Together, these studies make the evaluator part of the object being audited.

The present action-deletion problem is related but more specific. If a selected request node is masked, its action logit becomes unavailable. A large margin change can then occur even if the request features carried no useful evidence. A random baseline that removes more request nodes than the attention top-k is not a fair comparison. Type matching, selected-action protection, and clamp logging are introduced to control this mechanism. A leave-one-out perturbation ranking acts as a positive control for comprehensiveness, while Gradient x Input provides a comparator that does not use attention coefficients.

The small graph creates a second limit. When k = 3 and only a few valid nodes exist, a random size-three set must overlap substantially with the explanation set. This compresses the possible difference between them. Reporting the exact expected overlap and a taxi-only rank correlation makes that limitation visible instead of treating every near-zero DEF as decisive evidence.

2.6 Age of Information and telemetry degradation

Age of Information (AoI) measures the time since the newest received update was generated (Kaul, Yates and Gruteser, 2012). It is widely used to study communication freshness and the effect of delayed state on estimation or control. The explanation question is different. A dispatch action and its attention map may remain available even when a vehicle reading has stopped updating. The operator can see a strong weight without knowing that its source is old.

This dissertation uses AoI only as a node-level freshness descriptor and as the weight inside WAMSN. The manipulated condition is the declared duration of an observation-layer outage triggered by tunnel entry. The design does not manipulate AoI independently, so it does not claim that a larger AoI value causes lower faithfulness. Instead, it asks whether stale exposure and attention-based explanation behavior provide consistent evidence under controlled outages.

2.7 Research gap and positioning

The reviewed work establishes four points: relational RL is appropriate for fleet dispatch; attention is easy to expose but disputed as an explanation; GNN faithfulness requires graph-aware perturbations; and stale state matters to operational control. What is not joined is the assurance question at their intersection.

Within the graph- and RL-explainability studies reviewed in Sections 2.2-2.5, no evaluation protocol was identified that combines all of the following:

- a graph-attention MARL action explained at decision level;
- paired clean and stale versions of the same operational observation;
- explicit measurement of attention placed on stale telemetry;
- a control for graph nodes that also remove available actions; and
- replication across independently trained checkpoints.

The contribution is therefore an audit protocol and an empirical test, not a new dispatch algorithm. SUMO supplies a controlled simulated state (Lopez et al., 2018). The GAT-MAPPO policy supplies a realistic attention channel. The research contribution is to test whether that channel gives a reproducible freshness-aware and decision-relevant explanation, and to state clearly when the evidence cannot support that claim.

Chapter 3: Methodology

3.1 Study design

The study uses a controlled simulation because the research question requires the same decision to be observed in two ways: once with current telemetry and once with degraded telemetry. A real operational log cannot provide this exact clean twin. SUMO provides the simulated physical state, while a separate observation layer controls what the policy receives.

The protocol is defined in version-controlled configuration files. It contains three stages: training, validation-based checkpoint selection, and held-out evaluation. Faithfulness sweeps are run only after the selected checkpoints are frozen.

3.2 SUMO environment and data

The road network is an OpenStreetMap extract of the Central Park area in Yubei, Chongqing. Roads marked as tunnels supply physically meaningful signal-loss triggers. SUMO simulates 20 taxis and 50 passenger requests during each 1,200-second episode. Taxi dispatch is controlled through TraCI, while SUMO continues to update the true position and speed of every vehicle.

Demand variants are separated into training, validation, and test splits. The policy is fitted on training demand. Candidate checkpoints are compared on validation demand. The final reported runs use held-out test demand. This separation prevents test results from influencing model selection.

SUMO is a simulator rather than a neutral source of observed data (Lopez et al., 2018). Its outputs depend on the chosen road network, generated demand, routes, vehicle behavior parameters, tunnel locations, and random seeds. The experiment is therefore not described as free from bias. Instead, it uses these assumptions consistently in a paired design. At each sampled decision, the clean and degraded observations share the same underlying SUMO state and trained-policy checkpoint; only the observation-layer telemetry differs. This controls the comparison well enough to isolate the effect of the implemented degradation within this scenario, but it does not make the simulated fleet representative of every real city.

3.3 Observation graph and action space

Each idle taxi receives a self-centered graph containing:

Table 3.1: Observation graph node types and roles

Node type	Main features	Role
Self taxi	position, time, speed, AoI, position-valid flag	acting vehicle
Peer taxi	relative position, availability, distance, AoI	nearby fleet context
Passenger request	pickup/drop-off vectors, waiting time	candidate request and action

The graph contains up to five nearby taxis and five nearby requests. Features are normalized and request/taxi masks prevent padded nodes from receiving attention.

The discrete action space contains no-op plus one action for each visible passenger request. This one-to-one relationship is important: deleting a passenger-request node also removes the associated action, whereas deleting a peer-taxi node only hides information about that taxi. Section 3.7 explains how the faithfulness audit controls this asymmetry.

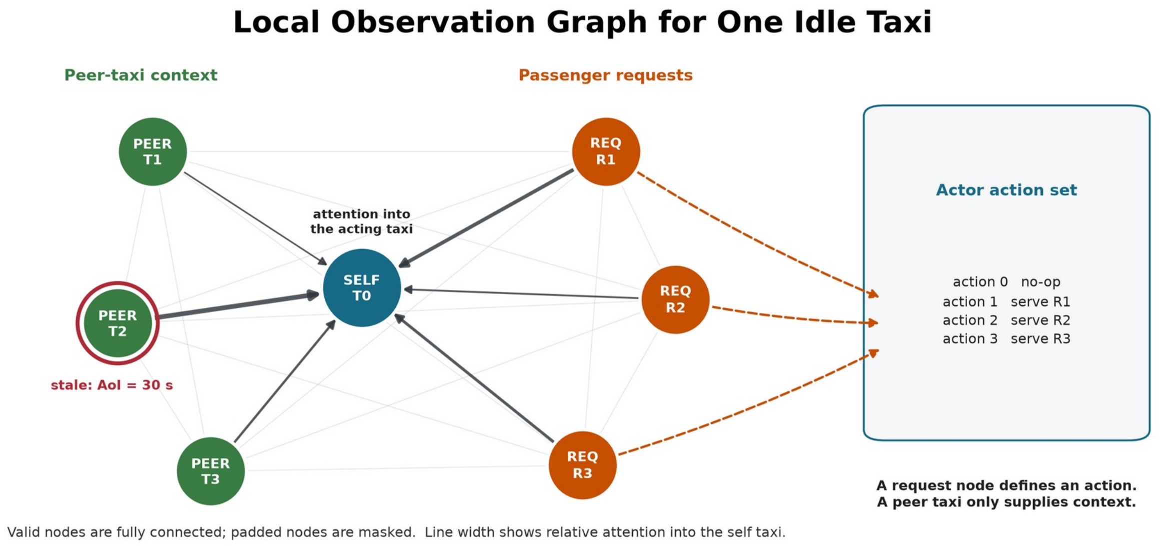

Figure 3.1. Valid self, peer-taxi and request nodes form a fully connected local graph. The red ring marks a stale peer observation, and the illustrative edge widths show attention into the acting taxi. Only request nodes map directly to dispatch actions, which motivates the construct-validity controls.

3.4 Policy models and training

The main policy uses per-node-type feature projections followed by two graph attention layers with four heads. The actor scores no-op and the visible requests. The critic estimates value for MAPPO training with centralized training and decentralized execution. The GAT implementation returns its attention tensors directly so they can be audited without changing the trained network.

The implementation uses scaled dot-product graph attention on the fully connected set of valid local nodes. This differs from the additive attention score in the original GAT paper, so "GAT" in the experiment labels denotes the graph-attention policy family, not an exact reproduction of that layer. For node i, node j, layer l, and head h:

```
q_i = W_Q,h LayerNorm(h_i^l),  k_j = W_K,h LayerNorm(h_j^l)
v_j = W_V,h LayerNorm(h_j^l)
s_ij^(l,h) = (q_i dot k_j) / sqrt(d_head)
alpha_ij^(l,h) = softmax_j(s_ij^(l,h))
z_i = concat_h sum_j alpha_ij^(l,h) v_j
u_i = h_i^l + W_O z_i
h_i^(l+1) = u_i + FFN(LayerNorm(u_i))
```

Here, FFN denotes the feed-forward network within each attention layer. The mask removes padded nodes before the softmax. A residual connection keeps the previous node embedding. The actor reads the updated self embedding for no-op and each updated request embedding for its matching dispatch action. The default explanation is the self row (i = 0) averaged over both layers and all four heads, then renormalized:

```
alpha_j = normalize((1 / (L H)) sum_l sum_h alpha_0j^(l,h))
```

This aggregation is declared before evaluation. A sensitivity analysis also reports each layer, head-wise values, head maximums, and attention rollout; it does not select an aggregation after seeing which result is favorable. The explanation average is an analysis choice: inside the policy, head outputs are concatenated and projected rather than averaged.

The fixed self row represents the acting taxi's dashboard view and is the explanation channel originally declared for audit. It directly contributes to the no-op embedding. A request-action logit, however, is read from that request's updated embedding, so its corresponding request row is more directly aligned with the selected logit. A sensitivity analysis therefore reports a separate request-action sensitivity comparison between the fixed self row and the selected request row. This comparison is not used to redefine the primary metric after seeing the result.

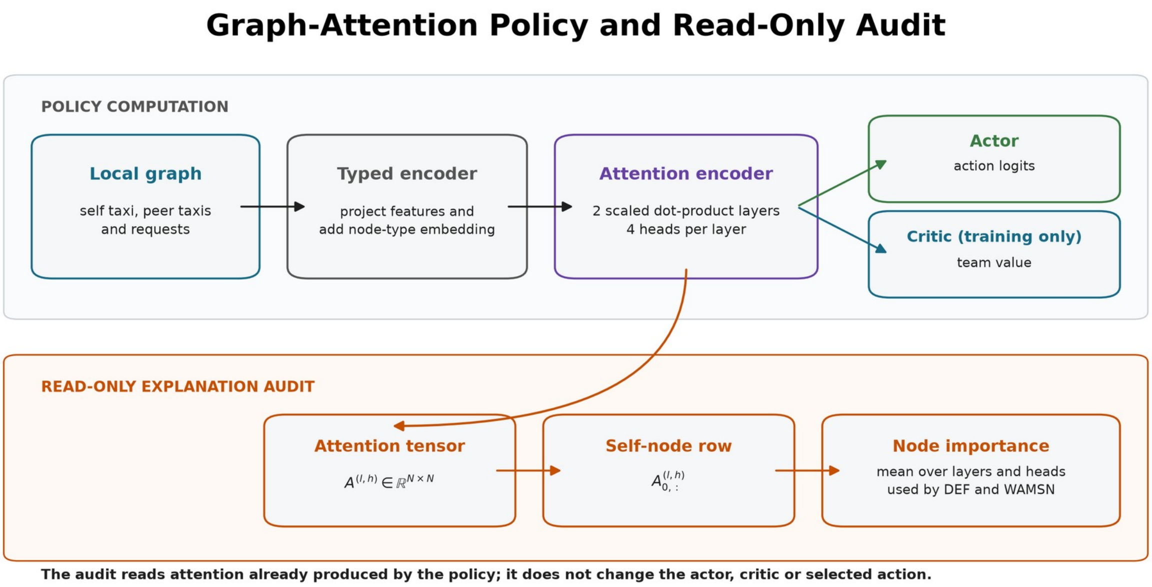

Figure 3.2. The policy and audit share the same two-layer attention encoder. The audit reduces the self-node attention row to node importance without modifying the actor, critic, or selected action.

Three policy conditions are included:

Table 3.2: Policy conditions used in the experiment

Model	Policy	Training observations	Purpose
MLP	MLP-MAPPO	clean	performance context without graph attention
GAT	GAT-MAPPO	clean	primary attention model under audit
GAT-Outage	GAT-MAPPO	tunnel-triggered 30-second outages	degradation-aware training condition

MLP is trained for 50 epochs, GAT for 40 epochs, and GAT-Outage for 50 epochs. Each condition uses independent training seeds 42, 43, and 44.

Candidate checkpoints include periodic snapshots saved every ten epochs, the final snapshot, and a rolling-best snapshot updated when the ten-epoch mean pickup count improves. The implementation records zero-based epoch indices, so index 39 denotes the checkpoint saved after 40 training epochs. Selection uses eight stochastic validation episodes with seed 2026 and mean pickups as the primary criterion; mean reward and the earlier checkpoint index break ties. The test split is not read during selection. The selected checkpoint indices are:

Table 3.3: Validation-selected checkpoint indices (zero-based)

Model	seed 42	seed 43	seed 44
MLP	20	40	34
GAT	39	39	30
GAT-Outage	49	40	40

The shared optimization settings are shown below.

Table 3.4: Shared training and optimization settings

Setting	Value
Training epochs	40 (GAT), 50 (MLP and GAT-Outage)
Learning rate	0.0001 (GAT variants), 0.0003 (MLP)
Discount factor (gamma)	0.99
Generalized advantage estimation factor (lambda)	0.95
PPO clip ratio	0.20
PPO passes per update	4
Minibatch size	256
Value-loss coefficient	0.25
Entropy coefficient	0.05, adaptive to a 0.20 entropy floor
Target Kullback-Leibler divergence	0.015
Gradient-norm limit	0.50
GAT hidden width / layers / heads	64 / 2 / 4

At simulation step t, the shared team reward is:

```
r_t = 10 N_pickup,t + 0.5 N_dispatch,t - 0.001 mean_wait_t
```

The reward trains dispatch behavior. It contains no term that rewards a faithful, stable, or human-readable attention map.

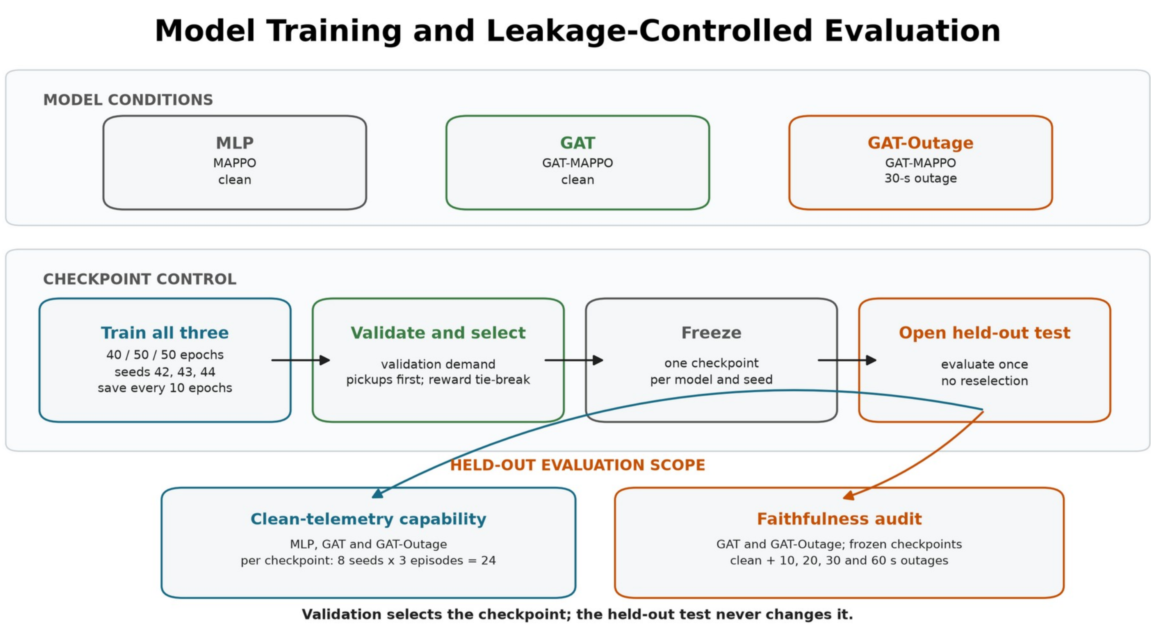

Figure 3.3. Common training and validation-based checkpoint selection for all three model conditions. Each frozen checkpoint receives 24 clean held-out episodes. The faithfulness audit then compares the frozen GAT and GAT-Outage checkpoints across clean and four outage conditions without reselection.

3.5 Telemetry degradation

The trigger and the degradation location are separate parts of the design.

Step 1 - Trigger: when a taxi enters a tunnel edge, the event starts one outage.

Step 2 - Observation-layer outage: for the declared duration, the policy sees the taxi's last valid position and speed. SUMO continues to simulate the true state.

Step 3 - Recovery: after the fixed window expires, a current reading is accepted. A taxi must leave and enter a tunnel again before another tunnel-entry event can trigger.

The fixed outage durations are 10, 20, 30, and 60 seconds. They are easy to monitor and compare, and they create increasing empirical degradation rates. They are not treated as direct causal doses of explanation faithfulness.

AoI is calculated as the current simulation time minus the time of the last valid update and is included in each GAT observation. WAMSN uses normalized AoI as a staleness weight. For this reason, an increase in WAMSN with outage duration partly verifies that the manipulation created longer stale exposure; it is not by itself proof that AoI changed DEF.

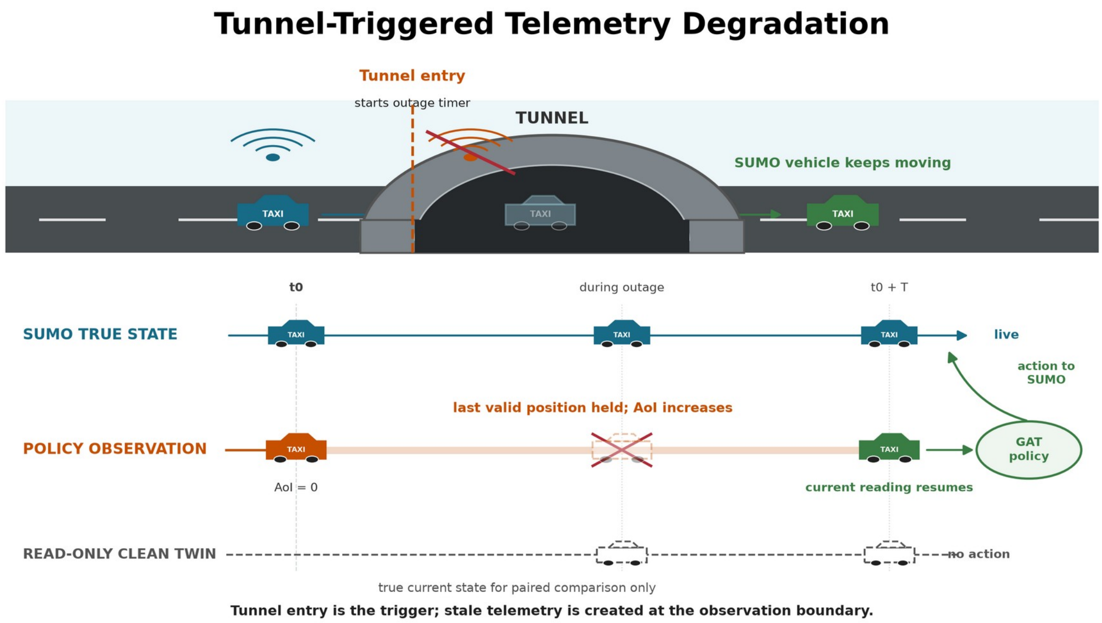

Figure 3.4. Tunnel entry triggers an observation-layer outage. SUMO keeps moving the vehicle while the policy sees its last valid position; the clean twin reads current state but never acts.

3.6 Explanation measures

3.6.1 Decision-level explanation faithfulness

DEF asks whether the nodes ranked highly by an explanation affect the chosen action more than a size- and type-matched random set. Two score functions are recorded from the same counterfactual forward passes. **Probability DEF** uses the selected action probability, p(G,a), and is bounded by the probability scale. **Logit-margin DEF** uses the selected action logit minus the strongest available alternative, m(G,a). The margin can retain resolution when the softmax probability is saturated, but its value depends on a checkpoint's logit scale.

For top-k nodes R_k, comprehensiveness measures the output change when those nodes are removed; sufficiency measures the output retained when only those nodes remain. The two gains are compared with five matched random subsets for each k in {1,2,3} and then averaged.

Let s(G,a) denote either p(G,a) or m(G,a). For explanation subset R_k and a matched random subset B_k:

```
Comp(R_k) = s(G,a) - s(G without R_k,a)
Suff(R_k) = s(G,a) - s(G keeping only R_k,a)
g_comp(k) = Comp(R_k) - mean_b Comp(B_k,b)
g_suff(k) = mean_b Suff(B_k,b) - Suff(R_k)
DEF = mean_k 0.5 [g_comp(k) + g_suff(k)]
```

Comprehensiveness is better when removing the explanation weakens the action more than removing random nodes. Sufficiency is better when keeping the explanation loses less evidence than keeping random nodes. The logit margin is clipped to [-10, 10] only when an action becomes unavailable or unopposed; every clamp is logged.

Type-matched random subsets are the **control baseline**; they are not themselves a faithfulness score. Corrected DEF is the measured difference between the attention-ranked subset and this matched random control. Its sign is interpreted as follows:

1. **Positive DEF:** the attention-ranked nodes affect the selected action more than comparable random nodes. This supports decision-level faithfulness.
2. **DEF near zero:** the attention ranking has no measured advantage over the matched random control.
3. **Negative DEF:** the attention-ranked nodes affect the selected action less than the matched random nodes. This does not support the attention ranking as a faithful explanation.

These interpretations apply to both variants. The confirmatory H1-H4 family and the paired clean-twin comparison use probability DEF, following the declared hypothesis analysis. Logit-margin DEF is used for the construct-validity and ranking-control diagnostics because it can show output movement that a saturated probability hides. It is action-protected where request deletion could remove the chosen action. The two variants have different units and cannot be converted into one another. Both use the type-matched baseline described in Section 3.7. DEF provides comparative perturbation evidence; even a positive value does not prove that attention is a complete causal explanation.

3.6.2 Stale-node attention

WAMSN measures attention attached to stale vehicle information:

```
WAMSN = sum(attention_i * normalized_AoI_i) / sum(attention_i)
```

Only self and peer-taxi nodes contribute because request nodes do not carry vehicle telemetry. The analysis reports WAMSN when at least one stale vehicle node is visible, together with stale-node attention share, stale-node mass, and whether a stale node enters the top three.

The paired stale-attention shift compares the degraded observation with its exact clean twin at the same simulation step. Positive values mean the degraded observation assigns more attention mass to nodes marked stale; negative values mean it assigns less.

3.7 Construct-validity audit

The audit supports the primary research problem by checking whether DEF is a valid comparison in this action-candidate graph.

A uniform random occlusion can select passenger requests more often than the attention top-k. Removing such a request changes both the observation and the available action set. A nearby-taxi occlusion changes only information. The result can therefore reflect different node-type composition rather than explanation quality.

The corrected protocol uses three controls:

- **type matching:** random subsets contain the same mixture of taxi and request nodes as the explanation subset;
- **chosen-action protection:** the request associated with the selected action is not deleted in the action-protected variant;
- **clamp logging:** counterfactual action-margin clamps are counted so that action deletion can be detected.

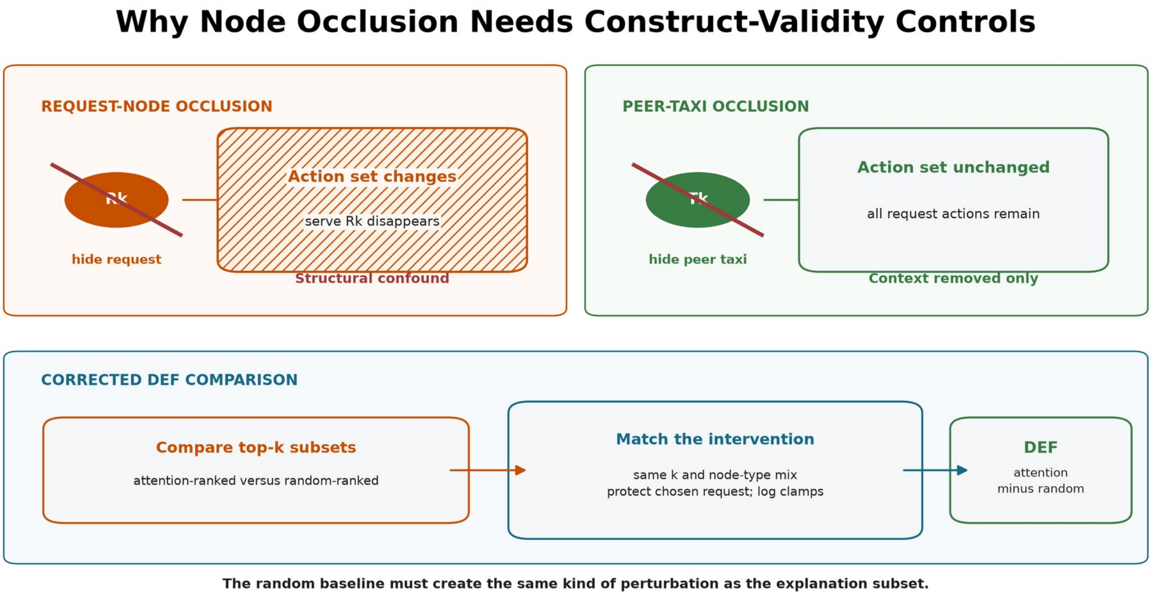

Figure 3.5. Request-node occlusion can remove its action, while peer-taxi occlusion removes context only. Corrected DEF matches subset size and node types, protects the chosen request, and logs margin clamps.

Uniform-baseline results are retained only as an audit trail. They are not used for the final hypothesis verdicts.

Three additional diagnostics test whether a near-zero DEF can be interpreted. First, a leave-one-out (LOO) control masks each valid node separately and ranks nodes by the resulting selected-action margin loss. The selected request is protected. This control is expected to strengthen comprehensiveness because it is built from the same single-node perturbation, but it is not an independent proof that the combined DEF is optimal. Second, Gradient x Input ranks each node by the L2 norm of the selected-action logit gradient multiplied by its raw features. It is a post-hoc comparator that does not use attention weights. Third, the audit reports valid taxi/request counts and the exact expected top-k overlap with a type-matched random subset. Taxi-only Spearman correlation between attention and LOO effects supplies a ranking measure that does not delete request actions and does not depend on random subset overlap.

3.8 Evaluation matrix

MLP, GAT, and GAT-Outage are evaluated under clean telemetry with eight evaluation seeds (42-49) and three episodes per seed. One checkpoint is frozen for each model and training seed, so this gives 24 held-out episodes per checkpoint. The clean capability evaluation therefore contains 216 episode evaluations:

```
3 models x 3 frozen checkpoints per model x 8 evaluation seeds x 3
episodes
```

The complete evaluation structure is summarized below. An episode evaluation is one complete rollout of one frozen checkpoint under one telemetry condition.

Table 3.5: Held-out evaluation structure and episode counts

Evaluation	No. of models	Frozen checkpoints	Telemetry cases	Episodes/cell	Total
Clean capability context	3	9	1	24	216
Exposure-conditioned faithfulness sweep	2	6	5	48	1,440
Added ranking controls	2	6	2	24	288
Random-trigger sensitivity analysis	2	6	3	24	432

GAT and GAT-Outage receive the full faithfulness sweep:

```
3 training seeds x 5 conditions x 8 evaluation seeds x 6 episodes
```

The five conditions and their roles are:

Table 3.6: Telemetry conditions used in held-out evaluation

Telemetry condition	Observation-layer treatment	GAT-Outage relation	Evaluation purpose
Clean	Current vehicle readings	No imposed outage	Capability context and faithfulness baseline
10 s	Freeze last valid reading for 10 seconds	Unseen shorter duration	Mild-degradation transfer test
20 s	Freeze last valid reading for 20 seconds	Unseen shorter duration	Intermediate transfer test
30 s	Freeze last valid reading for 30 seconds	Training-matched duration	In-condition degradation test
60 s	Freeze last valid reading for 60 seconds	Unseen longer duration	Severe stress and transfer test

The legal-random capability baseline samples uniformly from no-op and the currently valid request actions. The greedy baseline is deliberately naive: each idle taxi chooses its nearest visible request independently. If several taxis choose the same request, the environment accepts the first action in the step and rejects the later conflicts; there is no fleet-wide assignment or conflict optimization. The policy is deterministic for a fixed demand variant, so its uncertainty is based on three independent demand variants rather than the repeated evaluation-seed records. It is therefore interpreted only as a lower bound, not as a competitive dispatch algorithm.

The primary learned-policy evaluation samples from the policy distribution. This matches training and preserves both no-op and request-dispatch decisions for explanation auditing. A separate descriptive diagnostic evaluates each of the six selected GAT checkpoints by deterministic argmax over three held-out test episodes. This diagnostic is not used for checkpoint selection or the primary capability comparison. It checks whether sampled performance also corresponds to a usable deterministic policy.

The relation column refers only to GAT-Outage; the clean-trained GAT has not seen any imposed outage duration during training. The evaluation changes the observation-layer condition but never retrains or reselects a checkpoint. Outside stale exposure, faithfulness is sampled every 16 decisions. Every decision that contains at least one stale vehicle node is scored. Each such observation is also compared with its exact clean twin at the same simulation step. The degraded and clean evaluations reuse the same matched-random draw, so the paired difference does not contain avoidable baseline-sampling noise.

Each degraded condition must contain at least 20 stale-exposed records from at least three episodes. Preflight also checks expected cells, clean source provenance, type-matched controls, chosen-action exclusion, complete event capture, and separation in empirical AoI between conditions. All six sweeps pass these gates.

3.9 Hypotheses and statistics

The confirmatory hypotheses are:

- **H1:** DEF decreases as outage duration increases.
- **H2:** faithfulness declines faster than dispatch performance.
- **H3:** WAMSN increases as outage duration increases.
- **H4:** decisions with greater WAMSN have lower DEF within an episode.
- **H5:** compared with clean training, degradation-aware training reduces the decline in DEF associated with longer outages and higher WAMSN.

H1 and H3 use episode-block permutation trend tests and cluster bootstrap confidence intervals. H2 uses paired cell-level sign flips relative to each evaluation seed's clean condition. H4 calculates Spearman correlation within each episode before a sign-flip test. Holm correction is applied to H1-H4 within each trained policy.

The exposure-conditioned paired follow-up directly subtracts clean-twin DEF from degraded DEF for the same decision. Negative values mean lower measured faithfulness under the degraded observation. This paired estimate is reported with an episode-block confidence interval and is interpreted by effect size, direction across trained policies, and uncertainty rather than by a decision-level p-value alone.

H2 is retained as a pre-declared but exploratory comparison. Its original rate divides by clean DEF, which is close to zero and can make a small absolute change look arbitrarily large. The analysis therefore reports the support verdict and absolute trajectories, but does not use the ratio as the main evidence for faithfulness decoupling.

Thousands of decisions improve measurement precision but do not replace independent model replication. The final synthesis therefore treats each training seed as one trained-policy replicate. With only three seeds per model, the dissertation reports the range and number of supporting seeds and does not claim a cross-seed population p-value.

An exploratory action-stratified audit separates eligible decisions into no-op and dispatch. Decisions with no available request are excluded because no-op is then forced rather than chosen. Within each stratum, records are still averaged by episode before permutation tests or bootstrap intervals are formed. This diagnostic tests whether the combined statistics describe both actions or are driven by the more common action. It was added after the primary analysis and is interpreted descriptively rather than as a new confirmatory hypothesis.

A second sensitivity analysis checks whether the paired result is specific to the fixed tunnel location. The six checkpoints remain frozen and are evaluated under clean telemetry, a 30-second tunnel-triggered freeze, and a 30-second randomly triggered freeze. The random trigger is an independent Bernoulli draw for each taxi at each observation step. A preliminary run without faithfulness scoring selected a trigger probability of 0.0023: for the reference GAT checkpoint, this produced a 0.569% degraded-observation rate, close to the 0.580% tunnel rate. The same probability is then held fixed for all checkpoints. Because policy actions change taxi availability and trajectories, the realized exposure rates can differ; they are reported rather than assumed equal. The comparison is therefore interpreted mainly from paired shifts among decisions that actually contain a stale vehicle node.

The ranking controls use the same held-out demand and stochastic action protocol. They evaluate GAT and GAT-Outage under clean telemetry and a 60-second outage:

```
2 models x 3 training seeds x 2 conditions x 8 evaluation seeds x 3
episodes
```

LOO, Gradient x Input, overlap, and aggregation diagnostics are sampled every eight decisions. Query-row sensitivity is evaluated at every selected-request action because it asks whether the explanation row matches the node that produces that action logit. Decision records are averaged within episodes before bootstrap confidence intervals are calculated. Attention-aggregation sensitivity is reported per training seed and only for decisions where at least one stale vehicle node is visible.

3.10 Reproducibility

The experiment runner records the configuration, commands, seeds, checkpoint hashes, source revision, and preflight reports in versioned run directories. Compact outputs include the summary, performance context, deterministic diagnostic, action-stratified audit, and training-seed synthesis tables. The training-seed synthesis makes consistency across trained policies explicit. Reproduction commands, file locations, and expected outputs are documented in docs/REPRODUCE_EXPERIMENTS.md.

3.11 Ethics and data governance

The experiment uses a simulated taxi fleet, generated demand, and an OpenStreetMap-derived road network. It contains no human participants, personal records, live vehicle identifiers, or commercial dispatch data. Its main ethical risk is miscommunication: presenting attention as a trustworthy reason could give an operator false confidence. The reporting therefore keeps freshness, task behavior, and faithfulness separate and avoids unsupported causal claims.

Chapter 4: Results

The automated preflight audit passed 100 checks covering provenance and protocol requirements. Results are reported by training seed because decisions and episodes from one checkpoint do not constitute independent model replications.

4.1 Policy capability and training stability

Validation selected GAT checkpoint indices 39, 39, and 30 and GAT-Outage indices 49, 40, and 40 for seeds 42, 43, and 44. These are zero-based indices; for example, index 39 is the checkpoint saved after 40 training epochs. The selected index is reported separately for each trained policy.

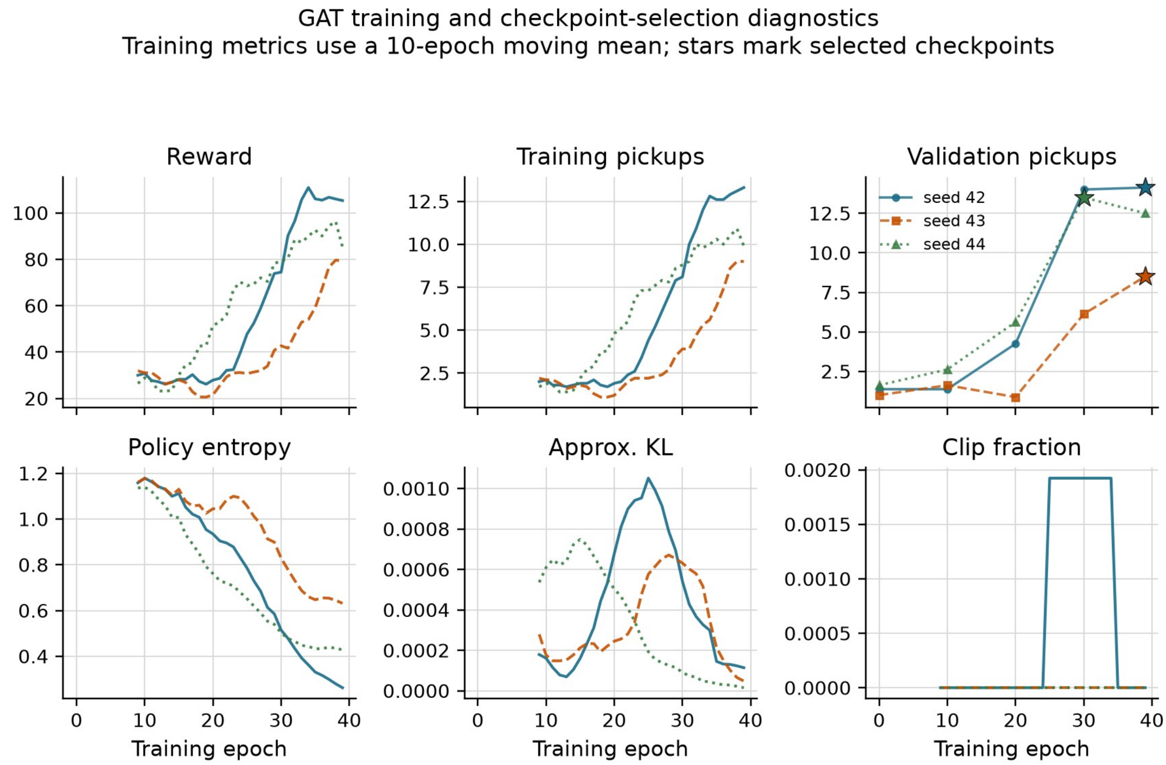

Figure 4.1. GAT training metrics use a ten-epoch moving mean. Stars mark validation-selected checkpoints. Reward and pickups still vary between epochs, but entropy remains above the collapse threshold and PPO clipping is limited.

All nine training runs pass the declared stability gates. The following values describe the final ten training epochs, not the held-out evaluation in Table 4.1. For GAT, the final ten-epoch mean pickups are 13.3, 9.0, and 9.9, and final validation retains 100%, 100%, and 92.6% of the selected validation score. The corresponding GAT-Outage values are 12.4, 14.1, and 13.8 pickups, with validation retention of 100%, 93.9%, and 96.5%. Training metrics still fluctuate across epochs, but no selected checkpoint is a zero-pickup policy under stochastic validation. This statement does not apply to deterministic argmax action selection, reported below.

Table 4.1 places the selected policies above two lower bounds evaluated on the same held-out demand. Performance is reported only to establish that the audited policies perform above random and greedy lower bounds.

Table 4.1: Policy capability on held-out demand

Policy	Mean pickups	Uncertainty or replicate range
Legal random	1.50	95% episode-cluster CI [1.08, 1.96]
Greedy nearest request	2.00	95% demand-variant CI [1.00, 3.00]
MLP	13.63	training-seed range 12.83-14.17
GAT	11.64	training-seed range 9.42-13.71
GAT-Outage	12.97	training-seed range 11.79-14.58

The legal-random baseline has 24 independent episode clusters. Greedy nearest is deterministic, so its effective sample is the three demand variants rather than 24 repeated seed-episode records. GAT's seed means are 13.71, 9.42, and 11.79. Every selected policy exceeds both lower bounds under stochastic action sampling. This establishes capability for the sampled policy distributions; it does not show that graph attention caused that capability or that the argmax policy is usable.

The audit does not require GAT to outperform MLP; it requires the audited checkpoint's sampled policy distribution to perform above the declared lower bounds. That condition is met. However, MLP has the highest cross-seed mean in this implementation, and the replicate ranges overlap. The experiment therefore does not establish a task-performance advantage for graph attention. This limits how broadly the audit can be generalized, but it does not answer the separate question of whether the exposed attention weights faithfully explain the GAT policies' decisions.

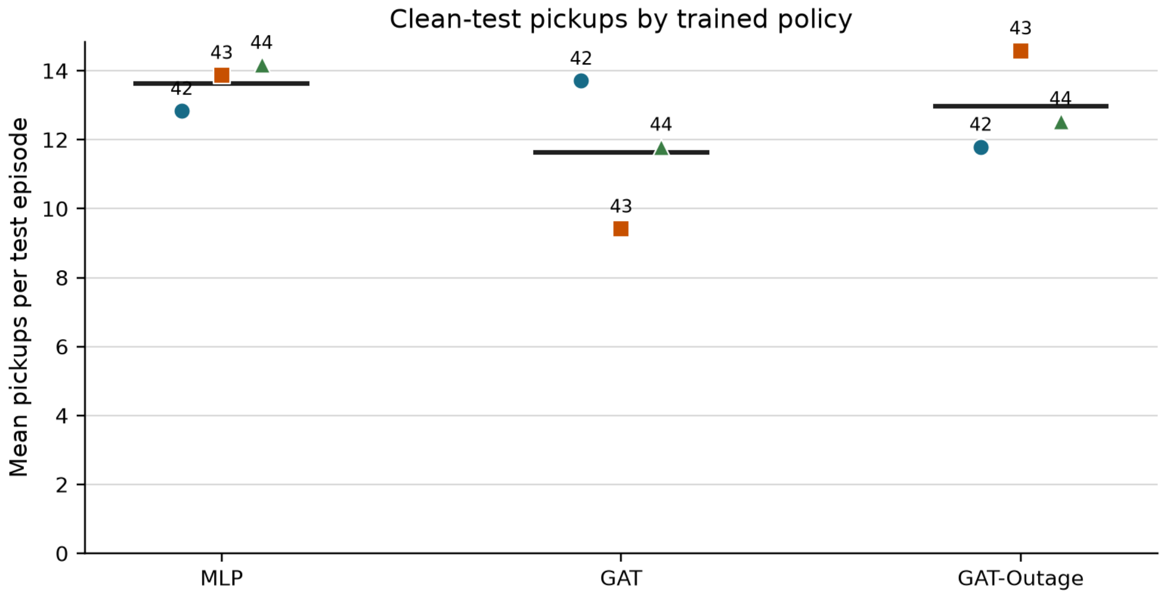

Figure 4.2. Each point is one selected policy evaluated over 24 held-out episodes. Horizontal lines show the cross-seed mean.

The deterministic diagnostic changes the capability interpretation. All six selected GAT checkpoints choose no-op throughout three held-out episodes, giving 0 pickups in 18 of 18 checkpoint-episodes. Their common mean reward is -46.58 and final pending-request wait is 905.9 seconds. The stochastic results are therefore not evidence of a stable argmax dispatcher. They show that the policy distributions retain enough probability on dispatch actions to produce pickups when sampled.

4.2 Construct-validity audit

Removing a request node can remove an available action, while removing a peer taxi only hides information. A uniform random occlusion therefore compares unlike interventions.

Table 4.2: Construct-validity audit of random controls

Control baseline used to calculate DEF	Mean margin-DEF	95% CI	Interpretation
Uniform random	-0.5540	[-0.5808, -0.5288]	dominated by node-type and action-deletion imbalance
Type-matched random	-0.0093	[-0.0159, -0.0013]	small negative result after matching node types

The type-matched interval excludes zero, but its magnitude is about two orders smaller than the uniform-control artifact. This 1,199-decision audit supports the measurement design; it is not a separate answer to the primary research question. The reported protocol uses type matching, protects the chosen request in the margin variant, and logs action-margin clamps.

4.3 Evaluator sensitivity and ranking controls

Raw attention, Gradient x Input, and an LOO perturbation ranking are evaluated under clean and 60-second conditions. Each model, training seed, and condition contains eight evaluation seeds and three episodes. Table 4.3 shows clean results. The 60-second aggregate means remain close to clean.

The two conditions are stored as separate evaluations and have different decision counts. Across the six checkpoints, the largest absolute 60-second minus clean change is 0.000071 for raw attention, 0.000254 for Gradient x Input, and 0.000306 for LOO. Figure 4.3 therefore shows the clean values and the condition difference separately instead of presenting two visually indistinguishable rows.

Table 4.3: Clean-telemetry faithfulness controls by training seed

Model	Seed	Raw DEF	GxI DEF	LOO DEF	Taxi-only rho
GAT	42	+0.0003	-0.0055	+0.0008	+0.088
GAT	43	-0.0009	-0.0033	+0.0099	+0.404
GAT	44	+0.0015	+0.0012	+0.0030	+0.426
GAT-Outage	42	+0.0011	-0.0019	+0.0020	+0.472
GAT-Outage	43	+0.0006	-0.0010	+0.0089	+0.602
GAT-Outage	44	+0.0031	+0.0052	+0.0044	+0.699

The LOO-ranked control is positive and larger than raw attention for all six selected checkpoints. This shows that DEF responds to a ranking built from single-node margin loss. LOO is still a perturbation control, not a ground-truth explanation. Gradient x Input changes sign across checkpoints and provides no uniform improvement.

Raw-attention DEF ranges from -0.0009 to +0.0031. Taxi-only rank correlation with LOO is positive in all six checkpoints, but ranges from 0.088 to 0.699. Attention therefore shows some agreement with decision-relevant LOO ordering, while the strength of that agreement remains checkpoint-dependent.

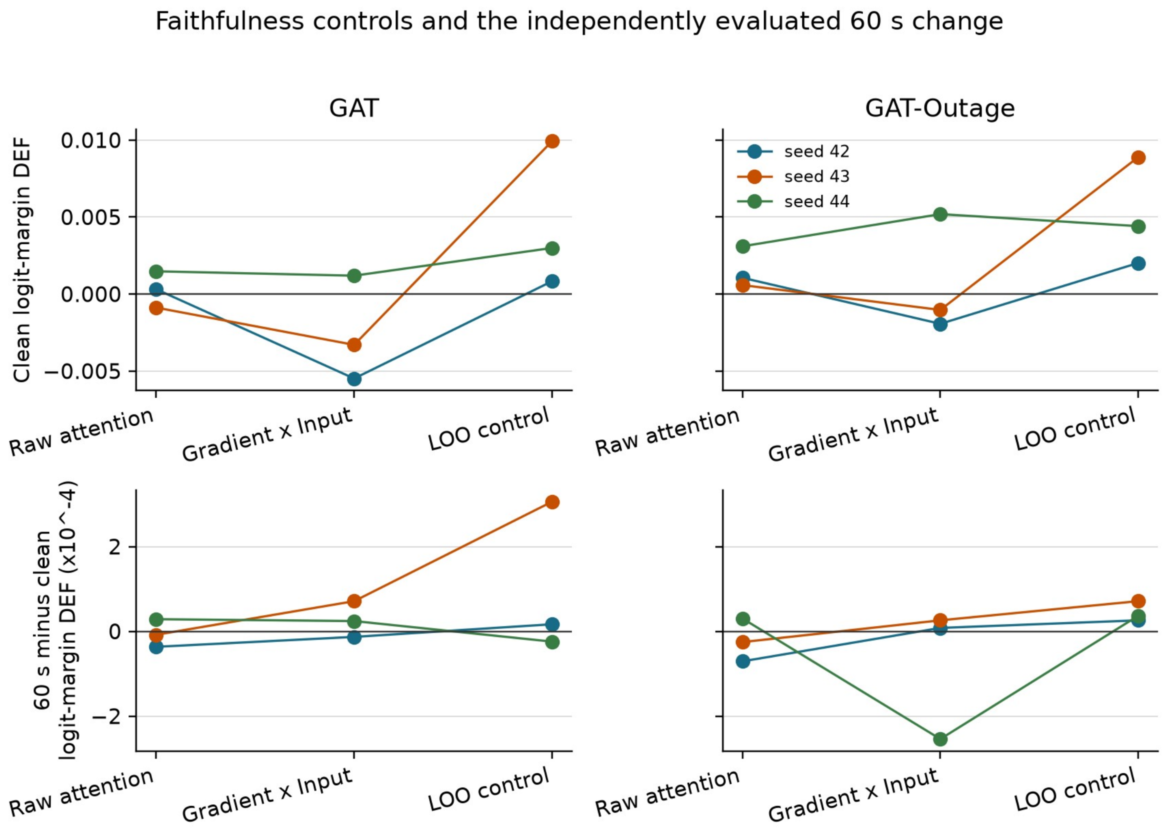

Figure 4.3. Top: clean logit-margin DEF. Bottom: 60-second minus clean DEF, multiplied by 10,000, from separate evaluations. Lines join methods for each checkpoint; non-zero changes confirm distinct records.

The attention reduction rule remains important. Across the declared mean, individual heads, layer maxima, and rollout, every checkpoint contains both a positive and a negative stale-attention shift. The widest ranges are -0.0109 to +0.0063 for GAT seed 44 and -0.0097 to +0.0064 for GAT-Outage seed 44.

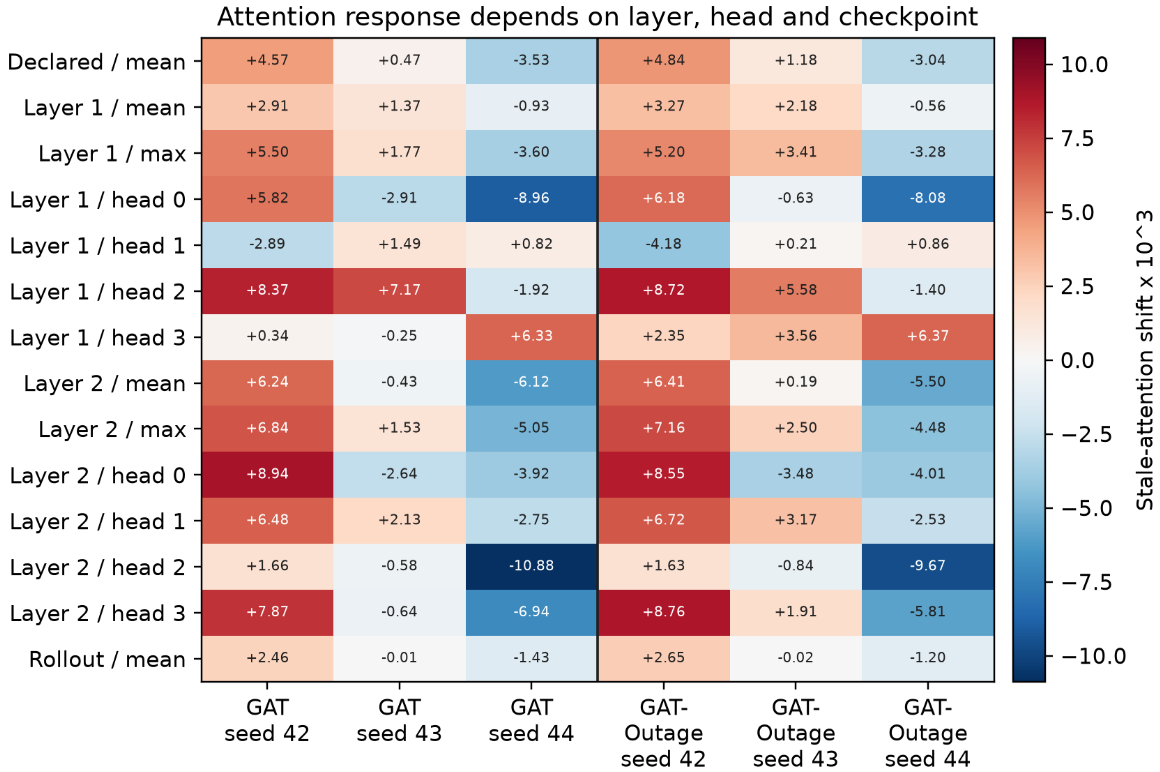

Figure 4.4. Stale-attention shift multiplied by 1,000. Selecting a layer, head, or rollout rule after seeing the result could reverse the conclusion.

Changing from the declared self row to the selected request row has little effect for seeds 42 and 44, but increases request-action DEF by about 0.0064 for GAT seed 43 and 0.0073 for GAT-Outage seed 43. The aggregate gap is +0.0022 for GAT and +0.0025 for GAT-Outage because it averages these different policy responses. The largest absolute 60-second minus clean query-row change is 0.000264, so the two conditions again remain close without being duplicates.

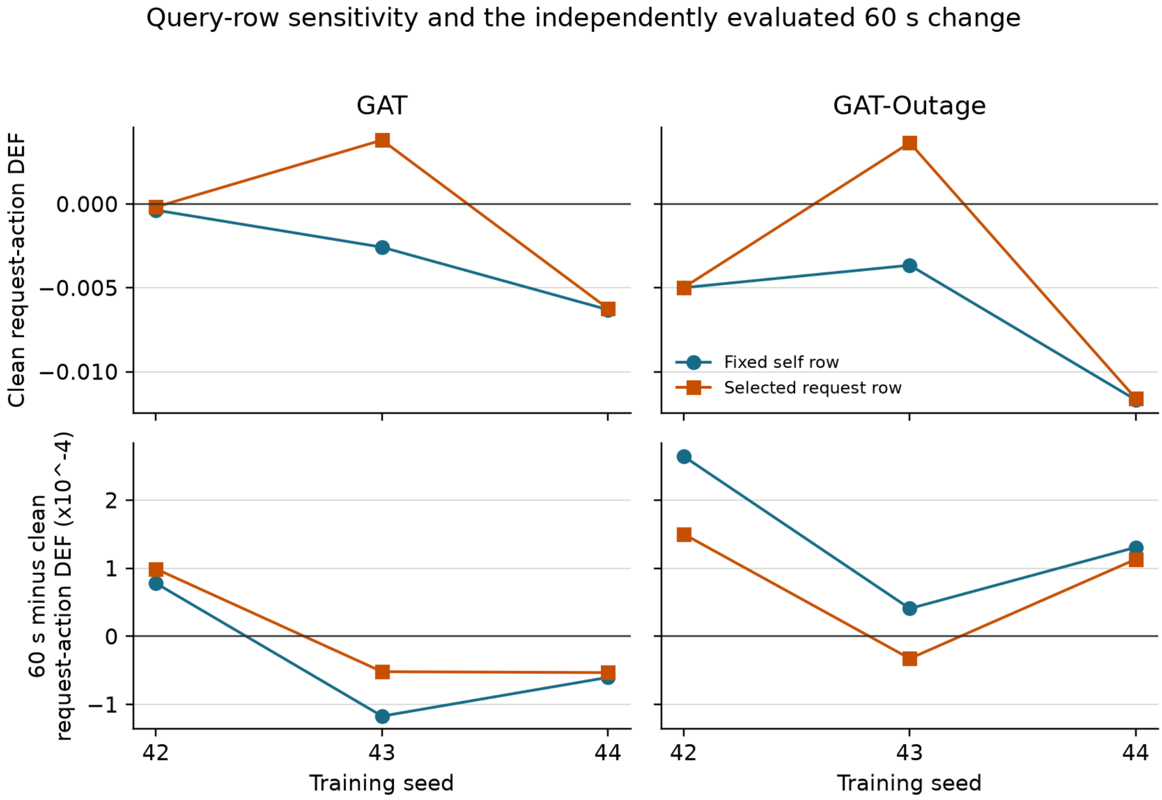

Figure 4.5. Top: clean request-action DEF from the declared self row and the selected request row. Bottom: each query row's independently evaluated 60-second result minus clean, multiplied by 10,000. The selected request row improves the seed-43 clean results but does not provide a consistent improvement across checkpoints or conditions.

4.4 Small-graph resolution

The scored graphs usually contain the maximum number of visible nodes.

Table 4.4: Valid node-count distribution in scored graphs

Model	Node count	Mean	Median	5th percentile	95th percentile
GAT	peer taxis	5.00	5	5	5
GAT	requests	4.51	5	1	5
GAT	non-self total	9.51	10	6	10
GAT-Outage	peer taxis	5.00	5	5	5
GAT-Outage	requests	4.52	5	1	5
GAT-Outage	non-self total	9.52	10	6	10

Even so, type matching creates substantial top-k overlap. The expected random intersection is about 0.22 for k=1, 0.41 for k=2, and 0.60 for k=3. Attention-LOO agreement at k=1 exceeds random overlap for both models. At k=3, it is 0.60 for GAT and 0.49 for GAT-Outage, compared with expected overlap of about 0.60. Restricting analysis to at least six non-self nodes does not change the sample because every scored decision already meets that threshold. This pattern suggests that the ranking agreement has more contrast at k=1; it does not establish that LOO is ground truth or that the top-ranked attention node is a causal explanation.

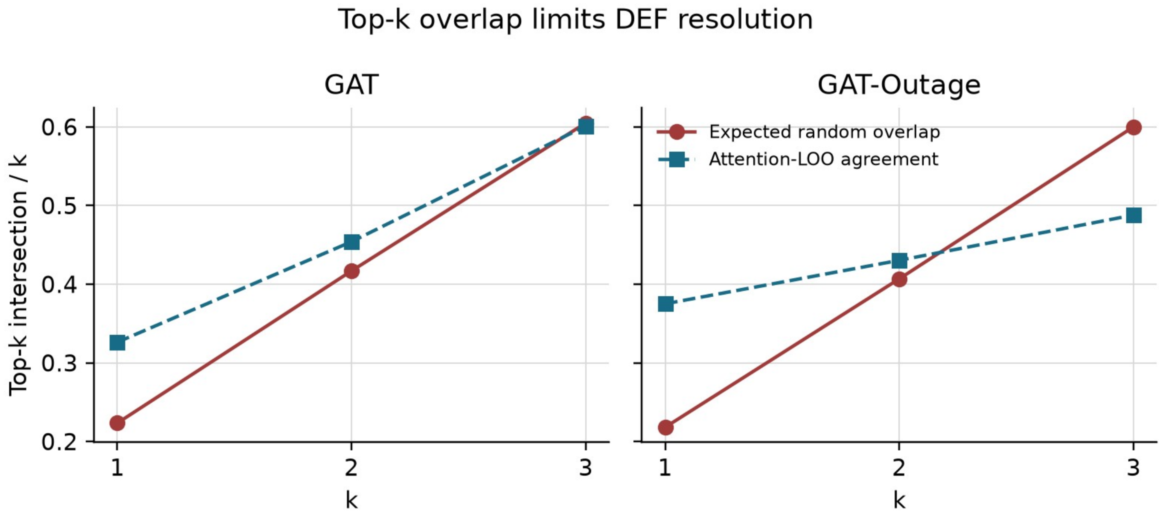

Figure 4.6. Large expected overlap between the explanation and its matched random set reduces the contrast available to DEF, especially at k=3.

4.5 Telemetry manipulation and stale exposure

Tunnel-triggered observation outages create increasing stale exposure. The event-aware audit scores every decision that contains at least one stale taxi, instead of relying on periodic sampling. Across GAT checkpoints, the number of stale-exposed records ranges from 347-382 at 10 seconds and 1,476-1,806 at 60 seconds. The GAT-Outage ranges are 353-372 and 1,566-1,697. Every degraded cell exceeds the pre-declared coverage gates.

Conditional WAMSN rises from 0.0275-0.0289 at 10 seconds to 0.0804-0.0900 at 60 seconds. H3 is supported for all six selected checkpoints. Because WAMSN contains normalized AoI, this verifies increasing stale exposure; it does not show that AoI causes lower faithfulness.

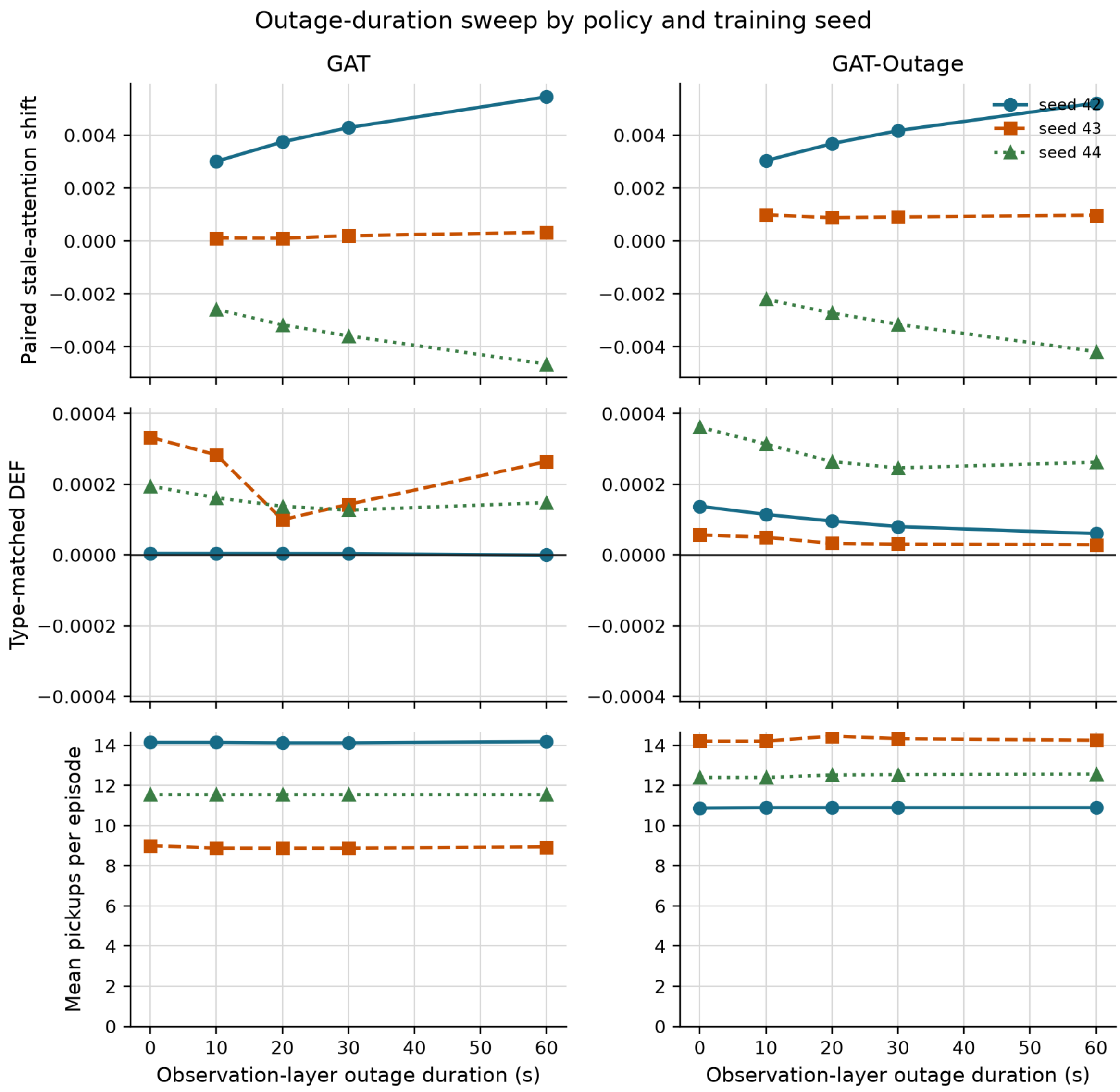

Figure 4.7. Longer observation outages increase WAMSN for every policy. DEF changes are much smaller, while pickups remain supporting context rather than the research outcome.

4.6 Exposure-conditioned paired audit

The paired audit compares each degraded observation with its exact clean twin at the same simulation step. Positive attention shift means more attention is assigned to nodes marked stale. Negative DEF shift means lower measured faithfulness under degradation.

Table 4.5: Exposure-conditioned paired attention and DEF shifts

Model	Seed	Attention shift [95% CI]	Paired probability-DEF shift [95% CI]
GAT	42	+0.004074 [+0.0039, +0.0042]	-0.0000041 [-0.0000060, -0.0000022]
GAT	43	+0.000208 [+0.0002, +0.0003]	-0.0000839 [-0.0001892, +0.0000143]
GAT	44	-0.003483 [-0.0036, -0.0034]	+0.0000092 [+0.0000068, +0.0000116]
GAT-Outage	42	+0.003975 [+0.0038, +0.0041]	-0.0000253 [-0.0000323, -0.0000193]
GAT-Outage	43	+0.000959 [+0.0009, +0.0010]	-0.0000114 [-0.0000343, +0.0000061]
GAT-Outage	44	-0.003038 [-0.0031, -0.0029]	+0.0000172 [+0.0000135, +0.0000212]

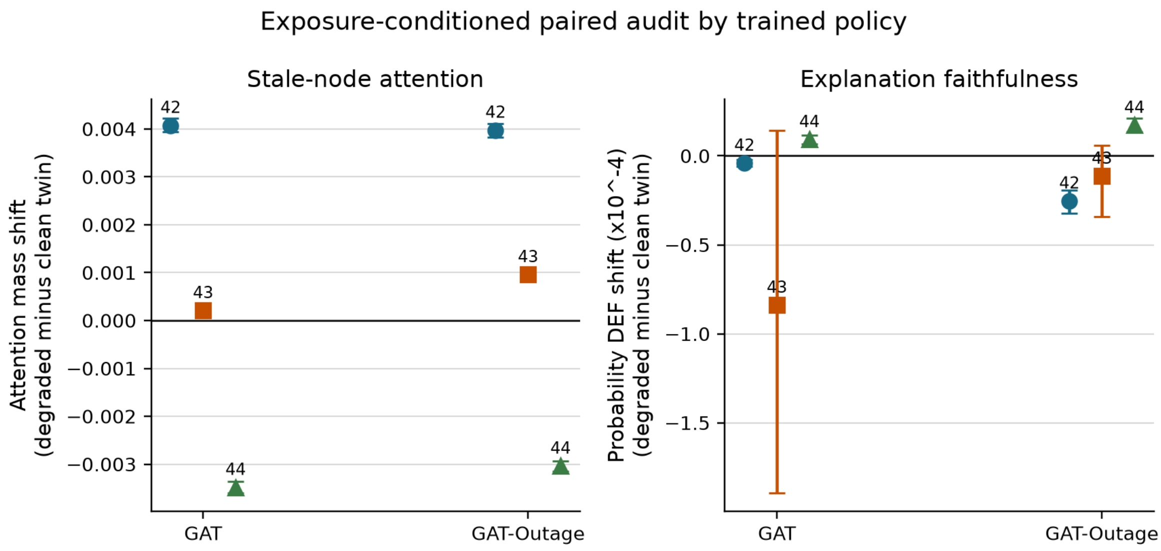

Figure 4.8. Checkpoints trained with seeds 42 and 43 move attention toward stale nodes and have a negative mean DEF shift. Checkpoints trained with seed 44 show the opposite directions. Error bars are 95% episode-block bootstrap intervals.

The same checkpoint pattern appears in both training regimes: four of six selected checkpoints have a positive attention shift and four have a negative mean DEF shift. Only the DEF decrease for checkpoints trained with seed 42 is separated from zero in both models; both seed-43 intervals include zero, while both checkpoints trained with seed 44 show a small increase. The effects are numerically small and directionally heterogeneous. This is not evidence for a universal claim that telemetry degradation moves attention toward stale nodes or always lowers faithfulness. The largest absolute paired probability-DEF shift is 0.0000839, equivalent to 0.00839 probability percentage points. Relative percentages are not reported because clean DEF is close to zero and would make such ratios unstable.

4.7 Action-stratified diagnostic

The combined faithfulness audit is dominated by chosen no-op actions. Across all six checkpoints, 56,550 records contain at least one valid request. Of these, 52,585 (93.0%) select no-op and 3,965 (7.0%) dispatch a request. Decisions with no valid request are excluded from these percentages.

Table 4.6: Action composition and dispatch-stratified DEF diagnostic

Checkpoint	No-op (%)	Dispatch (%)	Mean selected dispatch probability	Paired dispatch DEF shift [95% CI]
GAT, seed 42	96.6	3.4	0.0085	0.0000000 [0.0000000, 0.0000000]
GAT, seed 43	88.4	11.6	0.0368	-0.0020358 [-0.0032488, -0.0008371]
GAT, seed 44	90.4	9.6	0.0211	-0.0000011 [-0.0000029, 0.0000000]
GAT-Outage, seed 42	91.4	8.6	0.0228	+0.0000088 [+0.0000047, +0.0000131]
GAT-Outage, seed 43	94.6	5.4	0.0127	-0.0004403 [-0.0008368, -0.0001285]
GAT-Outage, seed 44	92.8	7.2	0.0181	-0.0000044 [-0.0000092, +0.0000004]

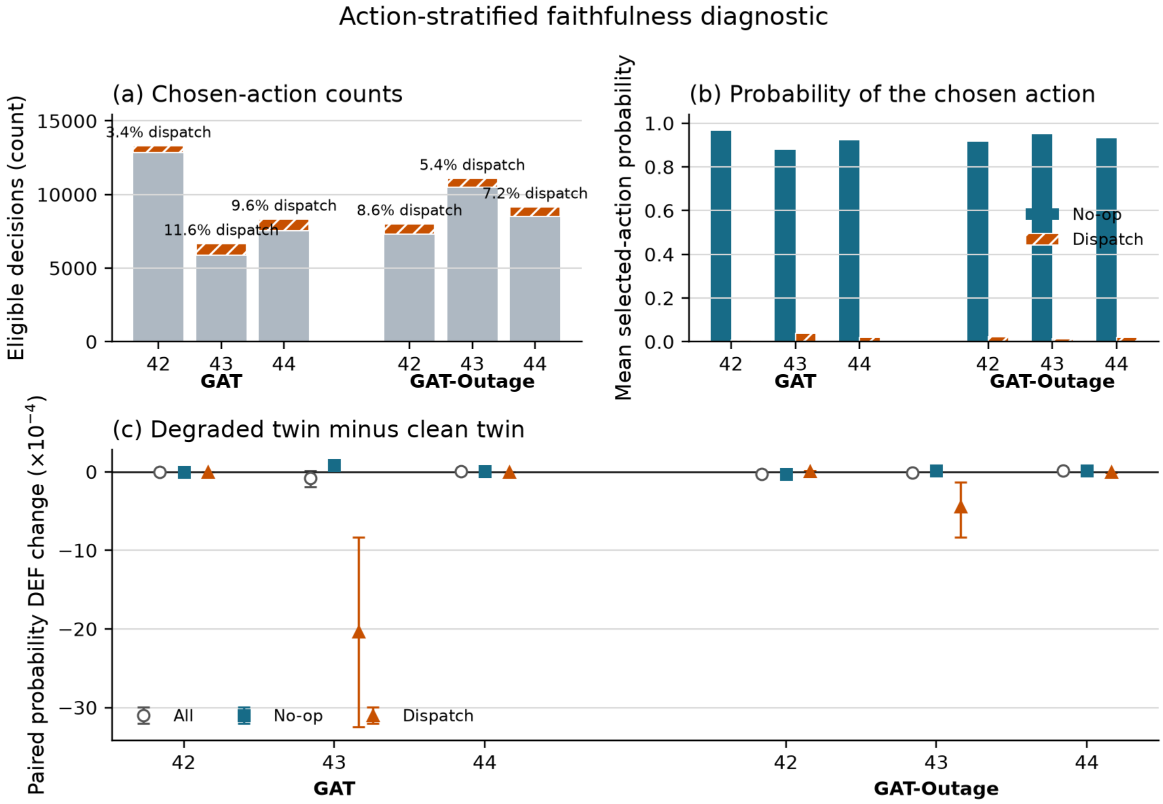

Figure 4.9. (a) Absolute no-op and dispatch counts; labels give the dispatch share. (b) Mean probability assigned to the action that was sampled. (c) Paired degraded-minus-clean probability DEF for all eligible decisions, chosen no-op actions, and dispatch actions. Error bars are 95% episode-block bootstrap intervals.

The no-op stratum closely follows the combined result because it supplies most records. The dispatch stratum does not reproduce the combined H1 trend in any checkpoint ($p = 0.44$-1.00), and it provides no consistent H4 evidence. The paired dispatch DEF shift is negative in four checkpoints, positive in one, and exactly zero in one. In four of six checkpoints its direction differs from the combined estimate. These effects are small except for the seed-43 checkpoints, and dispatch intervals are wider because fewer episodes contain scored dispatch actions. The diagnostic does not establish a separate law for dispatch decisions. It shows that the combined H1 and H4 findings should not be generalized to dispatch actions.

4.8 Random-trigger sensitivity analysis

The random-loss control replaces the fixed tunnel-entry trigger with an independent per-taxi trigger while retaining the same 30-second observation-layer freeze. The trigger probability is fixed across checkpoints, so realized exposure remains dependent on the trajectory induced by each policy. Random-to-tunnel exposure-rate ratios range from 0.44 to 0.98, with a median of 0.78. The comparison therefore focuses on the direction of paired effects within stale-exposed decisions rather than treating the two conditions as perfectly exposure matched.

Table 4.7: Tunnel-triggered and random-loss sensitivity results

Checkpoint	Exposure tunnel/random (%)	Attention shift tunnel/random (values x 1,000)	Probability DEF shift tunnel/random (values x 10,000)
GAT, seed 42	0.580 / 0.569	+4.281 / +4.037	-0.012 / +0.004
GAT, seed 43	1.176 / 0.517	+0.231 / -0.286	-0.479 / +7.361
GAT, seed 44	0.818 / 0.532	-3.623 / -3.165	+0.020 / +0.322

Checkpoint	Exposure tunnel/random (%)	Attention shift tunnel/random (values x 1,000)	Probability DEF shift tunnel/random (values x 10,000)
GAT-Outage, seed 42	0.967 / 0.713	+4.279 / +5.453	-0.205 / -0.093
GAT-Outage, seed 43	0.584 / 0.480	+0.958 / +0.444	-0.088 / -0.014
GAT-Outage, seed 44	0.712 / 0.648	-3.063 / -4.296	+0.091 / +0.518

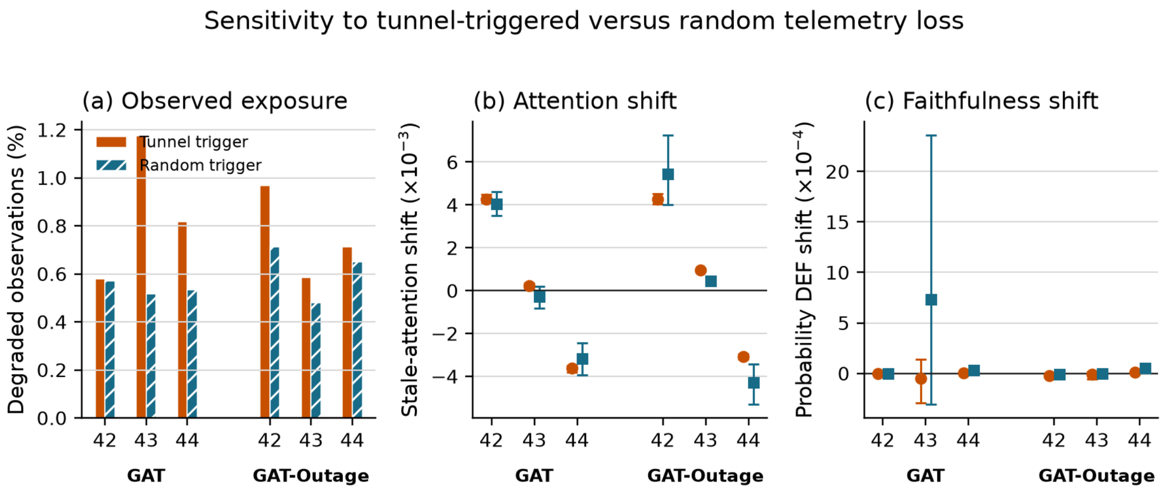

Figure 4.10. (a) Realized degraded-observation rates. (b) Change in attention mass on stale nodes relative to the exact clean twin. (c) Paired degraded-minus-clean probability DEF. Error bars are 95% episode-block bootstrap intervals. Both degradation conditions use a 30-second freeze.

Attention-shift direction agrees between tunnel and random triggers in five of six checkpoints. Probability-DEF direction agrees in four of six. One mismatch for GAT seed 42 consists of values very close to zero. GAT seed 43 is the clear exception, but its random-trigger DEF interval is wide and crosses zero. The seed-42 positive attention pattern and seed-44 negative pattern otherwise remain visible under random triggering in both training regimes. This reduces the likelihood that the overall checkpoint dependence is produced only by the chosen tunnel location. It does not establish a location-independent effect, because the realized exposure rates and stale-node populations are not identical.

4.9 Faithfulness hypotheses

The event-aware audit uses every decision with visible stale exposure rather than a periodic sample. H1 is supported for two of three GAT checkpoints and all three GAT-Outage checkpoints. H4 is supported for all six: within-episode WAMSN-DEF correlations range from -0.199 to -0.084 for GAT and from -0.423 to -0.212 for GAT-Outage. H3 remains consistently supported.

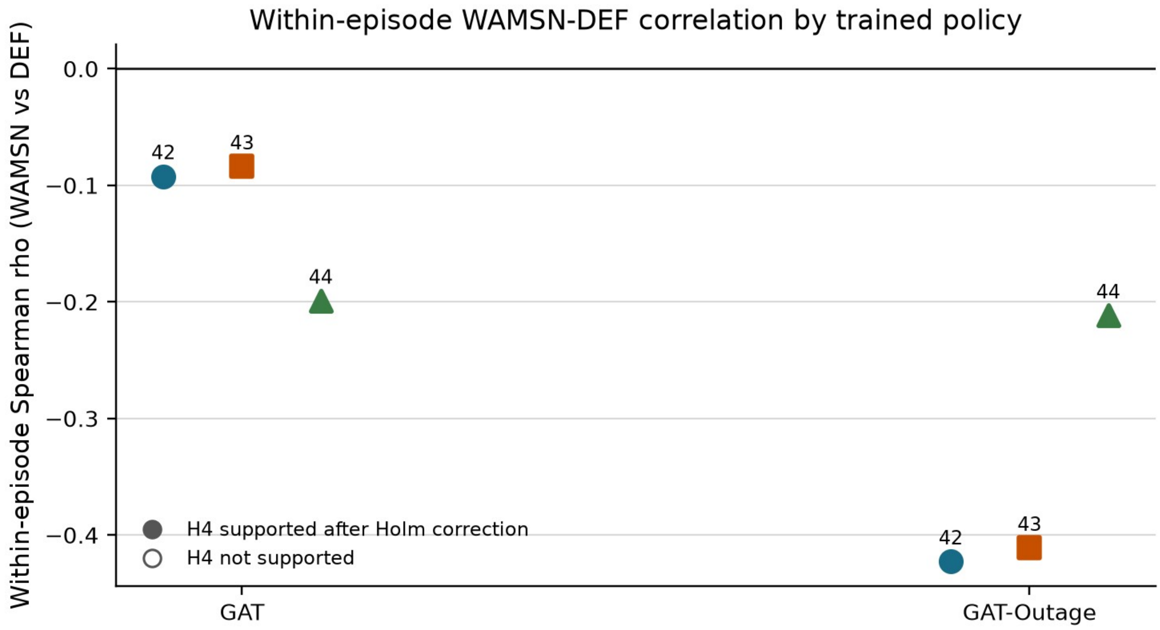

Figure 4.11. All checkpoints have a negative within-episode WAMSN-DEF association after Holm correction; the direct paired effect remains mixed.

Table 4.8: Hypothesis outcomes across training seeds

Hypothesis	GAT seeds supporting	GAT-Outage seeds supporting	Verdict
H1: outage duration increases, DEF decreases	2/3	3/3	strong but not universal
H2: faithfulness declines faster than performance	1/3	3/3	model-dependent; exploratory
H3: outage duration increases, WAMSN increases	3/3	3/3	consistently supported
H4: higher WAMSN is associated with lower DEF	3/3	3/3	consistently supported within episodes
H5: degradation-aware training reduces the adverse DEF relationships	n/a	0/3	not supported

H1 and H4 are association tests within one trained policy and primarily describe no-op decisions in this action-imbalanced sample. They do not by themselves establish that AoI causes lower faithfulness or that the same relationship holds for dispatch actions. The paired clean-twin audit supplies the more direct intervention contrast, and its sign still changes with training seed. H2 is retained as exploratory because clean DEF is close to zero and the relative-rate formulation is unstable.

4.10 Result summary

Longer outages reliably increase stale-data exposure, and greater WAMSN is associated with lower DEF within episodes. However, paired attention and DEF shifts reverse for seed 44, and 30-second outage training does not remove this checkpoint dependence. The combined results are dominated by no-op decisions, do not reproduce for dispatch actions, and retain exceptions under random triggering. Raw attention therefore provides no reproducible freshness or faithfulness guarantee across the independently trained policies tested here.

Chapter 5: Discussion

5.1 Answer to the central problem

The central problem is whether a complete graph-attention map can be trusted when some of its vehicle data are stale. The answer from this experiment is: **not without separate validation**. Raw attention is not shown to provide a stable assurance of freshness or decision relevance.

This answer does not depend on one p-value. Longer outages increase stale-data exposure, and within-policy tests usually associate that exposure with lower DEF. The direct clean-twin comparison is less uniform: attention reallocation and paired DEF shift have the expected directions for checkpoints trained with seeds 42 and 43, but both directions reverse for checkpoints trained with seed 44 under both training regimes. The measured effects are also small.

The action-stratified diagnostic prevents a broader interpretation. Chosen no-op actions account for 93.0% of eligible decisions, and the combined H1 and H4 patterns are not reproduced among dispatch actions. Four checkpoints also change paired-DEF direction between the combined and dispatch strata. The primary statistics therefore characterize the sampled action mixture, not a universal explanation response for both dispatch and no-op decisions.

The supporting controls make the interpretation stronger and narrower. The LOO-ranked control produces a positive DEF for every checkpoint, so the evaluator has some sensitivity to a perturbation-based ranking. However, expected top-k overlap reaches about 60% at k=3, which limits resolution. Taxi-only attention-LOO correlation is positive but varies substantially in strength. The result is not "the evaluator is perfect and attention fails." It is "within a measured but limited audit, attention gives no reproducible explanation guarantee."

Cross-seed evidence matrix

Question	Measure	Cross-seed result	Verdict
RQ1	Clean DEF	Near type-matched random baseline	**Not validated**
RQ2	Paired attention shift	Positive in 4/6 trained policies	**Mixed by seed**
RQ3	Paired DEF shift	Negative in 4/6 trained policies	**Mixed by seed**
RQ4	Degradation training	No consistent mitigation	**Not supported**

Conclusion: attention and paired faithfulness responses are not consistent across trained policies.

Figure 5.1. The evidence separates capability, evaluator validity, stale exposure, attention response, and decision relevance. These are not treated as one causal chain.

5.2 What the stale-exposure result means

WAMSN rises with outage duration for every GAT and GAT-Outage checkpoint. It is useful as an operational exposure indicator: it shows when an attention map contains more weight attached to old vehicle readings.

WAMSN is not evidence that AoI causes lower faithfulness. The metric itself weights attention by normalized AoI, so part of the increase follows directly from the manipulation. H1 and H4 show that DEF tends to be lower at greater exposure within most frozen policies. The paired audit asks a stricter question: does replacing the current observation with its degraded twin change attention and DEF at the same decision? That result changes sign across checkpoints trained with different seeds. Exposure, within-policy association, and paired intervention therefore answer different questions and should remain separate.

5.3 Dependence on checkpoint and analysis choice

The primary aggregation gives positive stale-attention shifts for checkpoints trained with seeds 42 and 43 and negative shifts for checkpoints trained with seed 44. The values are small: approximately -0.0035 to +0.0041 of total attention mass. Paired probability-DEF shifts are also small and reproduce the same direction in both training regimes. This shared pattern across the two model conditions suggests sensitivity to the trained checkpoint, not a stable response created by outage-aware training. The between-checkpoint range is larger than the observed difference between the two training-condition means. With only three training seeds, this is a descriptive comparison of selected checkpoints, not a variance decomposition or evidence that the seed value itself causes the response.

The random-trigger sensitivity analysis checks whether this pattern is only an artifact of the fixed tunnel location. Attention-shift direction is retained in five of six checkpoints, and probability-DEF direction in four of six. The seed-42 positive attention pattern and seed-44 negative pattern appear under both triggers and both training regimes. This makes a tunnel-only explanation less plausible, but it is not a complete location control. A fixed trigger probability produces different realized exposure rates after each policy changes taxi trajectories, and the identity of the stale taxis also changes. The result supports checkpoint dependence; it does not prove one general effect of random or tunnel-triggered loss.

The paired probability-DEF effects are small enough that their practical importance should not be overstated. Reading all six estimates as operationally close to zero is a reasonable challenge. The evidence against a dependable explanation is not a large universal decrease; it is that even these small changes reverse direction across independently trained checkpoints, while the raw clean-telemetry DEF is already near its matched control. The study therefore supports a lack of reproducible assurance, not a claim of large degradation damage.

The aggregation audit reveals a second problem. Individual heads can reverse the sign within the same checkpoint. The six checkpoint ranges extend from -0.0109 to +0.0089 across the tested reductions. An analyst could therefore obtain a different verbal conclusion by selecting another head or layer after seeing the data. The declared mean remains the primary result, but the sensitivity analysis shows that it is not a unique explanation produced by the network.

Query-row choice is similarly checkpoint-dependent. The selected request row improves request-action DEF by about 0.0064 for GAT seed 43 and 0.0073 for GAT-Outage seed 43, but changes little for the other four checkpoints. This does not support the claim that one query-row change generally corrects near-zero or negative DEF. An operator-facing attention map needs a declared and tested rule for choosing its query, layer, and heads.

The deterministic diagnostic reveals a related policy limitation. All six GAT checkpoints choose no-op in all 18 argmax episode evaluations, although their sampled policies outperform the declared lower bounds. This does not invalidate the faithfulness calculations for the actions actually sampled. It does mean that the experiment audits stochastic policy decisions and does not establish a deployable deterministic dispatcher. The assurance process should therefore report action composition and deterministic behavior before an attention map is exposed to an operator.

5.4 Role of the construct-validity audit

The construct-validity audit supports the primary experiment. Request nodes are both information and actions; peer taxis are context only. Uniform random deletion therefore creates a large action-removal artifact. Type matching and chosen-request protection make the comparison fairer.

The LOO control checks a different question: can DEF respond when nodes are ranked by their own single-node margin loss? Its consistently positive result shows limited sensitivity, mainly for comprehensiveness. It is not an oracle for the combined comprehensiveness-sufficiency score. Gradient x Input changes sign across checkpoints and is not automatically a stronger explanation in this task.

Together, these checks change the conclusion from a simple “attention is near random” statement to a more defensible one. Raw attention remains below the LOO control in all six checkpoints and has positive taxi-only agreement with LOO, but both DEF and rank agreement vary by trained policy. The aggregation and query-row audits add further sensitivity. This is partial evidence of decision relevance, not a stable default assurance.

The top-k diagnostic further limits this interpretation. Attention-LOO overlap is above the expected matched-random overlap at k=1, but it offers no clear advantage over that baseline at k=3. This is consistent with reduced resolution as larger subsets overlap, not proof that the top attention node is the model's true reason or that every three-node display is random.

5.5 Degradation-aware training

Training with 30-second outages does not solve the explanation problem. GAT-Outage retains the same seed-dependent attention and paired DEF directions as GAT. It strengthens the H1 and H4 associations, but does not make the direct degraded-minus-clean response consistent across independently trained policies.

This does not show that all robustness training is ineffective. The reward contains no term for explanation stability or faithfulness, so task-reward optimization alone is not designed to align attention with a human explanation. A future intervention would need an explicit explanation objective and evaluation on independently trained models.

5.6 Proposed assurance approach

The thesis responds to the identified risk with a **freshness-aware explanation assurance approach**. It does not change the trained policy or claim to make attention faithful. Instead, it provides a release process that reduces the risk of presenting an unsupported attention map as a trustworthy explanation. Table 5.1 links each observed risk to the required evidence and release rule.

Table 5.1: Freshness-aware explanation assurance approach

Observed risk	Assurance measure	Evidence used	Release rule
Attention may be attached to stale data	Report freshness separately	AoI and WAMSN	Never infer freshness from attention strength
Attention may not identify decision-relevant nodes	Use action-aware, type-matched tests	DEF, margin-DEF, and LOO	Do not claim a reason without evidence of decision relevance
Combined results may be dominated by no-op	Report no-op and dispatch strata	Action counts and stratified DEF	Do not generalize a combined result across action types
The result may depend on attention extraction	Test the query row, layer, head, and aggregation	Sensitivity audits	Predeclare the extraction rule and report material sensitivity
One checkpoint may not represent another	Audit each release checkpoint	Per-checkpoint results	Repeat after retraining or replacement
A pattern may not reproduce	Compare independent training runs	Cross-seed synthesis	Require a consistent conclusion across runs

The process is sequential: report freshness and action composition separately; run the action-aware audit; predeclare and test the attention extraction rule; audit every candidate checkpoint; and require a consistent conclusion across independent training runs. A failure at any stage keeps the attention map as an internal diagnostic.

Sampled dispatch performance provides capability context, but it cannot show that the displayed reason is current or decision-relevant.

This is a risk-control and validation solution, not a model-level cure for faithfulness decoupling. A model-level intervention would require an explicit explanation objective and independent evaluation, as discussed in Section 5.8.

5.7 Limitations

The study uses one simulated district, 20 taxis, and 50 requests. Tunnel exposure remains scenario-dependent, although the event-aware audit scores every observed stale-exposed decision. It evaluates three independent training seeds per model. Three seeds are sufficient to reveal heterogeneity, but not to estimate a population distribution of training outcomes. The selected policies pass the training stability gates and their stochastic evaluations outperform random and greedy lower bounds. However, every selected GAT checkpoint collapses to no-op under the small deterministic diagnostic. The study therefore does not establish deterministic deployment capability and does not isolate the contribution of graph edges to sampled capability.

The local graph is small. Type matching makes random top-k sets overlap heavily with the explanation, which compresses DEF. LOO is based on the same single-node perturbation as comprehensiveness and is not an independent ground truth explanation. Other perturbation choices or post-hoc explainers may give different results.

The degradation layer freezes position and speed for fixed windows. Real telemetry may include delayed packets, partial failures, map-matching errors, and asynchronous recovery. The study audits attention as an explanation; it does not claim that attention is useless inside the policy computation.

The paired clean/degraded comparison gives the study internal validity for its implemented observation-layer intervention: both observations are derived from the same simulated traffic state. It does not remove simulation or scenario-design bias. Road-network choice, generated demand, vehicle behavior, and tunnel placement may affect which taxis become stale and how those taxis are positioned in the local graph. In particular, a fixed tunnel can create an exposure-selection effect if its traffic conditions differ from the rest of the network. The conclusions are therefore limited to the tested SUMO setting and should not be read as population estimates for real fleets or cities.

5.8 Future work

Future experiments should add training seeds and use multiple road networks, demand levels, tunnel placements, and public mobility traces. The present random-trigger control should be extended to several calibrated exposure rates and persistent loss processes so that trigger location, exposure frequency, and outage duration can be separated. Experiments should also include disconnected-edge and message-passing ablations to establish how much the policy uses its graph channel. Delayed, intermittent, and biased telemetry should be compared with fixed freezes.

The explanation audit should also compare LOO with graph-specific post-hoc methods and evaluate objectives that directly reward explanation consistency. Any proposed improvement should be tested on held-out checkpoints, not only on more episodes from one trained model. A human-factors study could then test whether explicit freshness warnings lead operators to interpret the map more appropriately.

Chapter 6: Conclusion

This dissertation examined a practical assurance problem: a graph-attention map can remain visible and appear complete after some vehicle telemetry has become stale. Attention strength alone does not tell an operator whether its source is current or whether the highlighted node affected the action.

The central answer is clear: **this experiment does not validate raw graph-attention weights as dependable explanations under clean or degraded telemetry**. This is not a claim that every attention map is wrong. It means that attention cannot be assumed trustworthy by default.

The research questions lead to that answer:

- **RQ1: Is attention faithful under clean telemetry?** No stable result is found. Raw-attention DEF ranges from -0.0009 to +0.0031 across checkpoints. LOO gives a larger positive result for all six checkpoints, while taxi-only attention-LOO rank correlation ranges from +0.088 to +0.699.
- **RQ2: Does attention move toward stale vehicle nodes?** Not consistently. The declared aggregation moves toward stale nodes for checkpoints trained with seeds 42 and 43 and away from them for checkpoints trained with seed 44 under both training regimes. A random-trigger control retains the attention-shift direction in five of six checkpoints, while individual heads can still reverse the direction within one checkpoint.
- **RQ3: Does the explanation remain dependable as outage duration increases?** No cross-policy guarantee is found. H1 is supported in five of six selected checkpoints and H4 in all six, but the direct paired DEF shift is negative for four checkpoints and positive for two. The effect is small and reverses for the checkpoints trained with seed 44. Moreover, 93.0% of eligible decisions are no-op, and the H1/H4 patterns are not reproduced in the dispatch stratum.
- **RQ4: Does degradation-aware training improve explanation reliability?** No consistent mitigation is observed. GAT-Outage retains the same seed-dependent paired direction as GAT.

The construct-validity audit is essential to this interpretation. Deleting a request can also delete an action, while deleting a peer taxi only hides context. Type-matched and action-protected controls remove much of this artifact. The positive LOO result shows that DEF has some perturbation sensitivity, but the high expected top-k overlap limits its resolution. The conclusion is therefore based on several checks rather than one near-zero number.

The 30-second random-trigger sensitivity analysis reduces concern that the checkpoint-dependent pattern is produced only by the selected tunnel. It does not remove the need for multiple tunnel placements or real telemetry traces, because realized exposure and the stale-node population still differ between trigger mechanisms.

The capability evidence also has a clear boundary. Sampled policies outperform the basic lower bounds, but all six GAT checkpoints choose no-op in 18 of 18 deterministic diagnostic episodes. The study audits explanations for sampled stochastic actions; it does not demonstrate a deployable argmax dispatcher.

In this thesis, **faithfulness decoupling** is the assurance gap between a visible explanation and valid supporting evidence. An attention map can outlive the freshness of its inputs without giving a stable indication of decision relevance. The gap does not require DEF to decline monotonically with outage duration.

The thesis addresses this gap with a freshness-aware explanation assurance approach. It separates freshness and action composition from attention, uses action-aware and type-matched tests, checks the attention extraction rule, audits every candidate checkpoint, and requires consistency across independent training runs. Stable dispatch output is not an explanation test.

The approach controls operational risk; it does not claim to remove faithfulness decoupling from the model. Attention remains an internal diagnostic unless the full assurance process provides consistent evidence.

List of Publications

No publication is claimed in this dissertation.

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Appendices

Appendix A: Experiment Reproduction

The complete reproduction procedure is provided in docs/REPRODUCE_EXPERIMENTS.md. Compact citable outputs are stored in the version-controlled results directory identified in that guide.

Appendix B: Supporting Artefacts

The repository contains the experiment configuration, source code, validation selections, preflight reports, per-seed analyses, summary tables and figure sources used in this dissertation. Raw cells and checkpoints remain outside the document because of their size.